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Back to common sense: Oxford dictionary descriptive knowledge augmentation for aspect-based sentiment analysis

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ABSTRACT

Aspect-based Sentiment Analysis (ABSA) is a crucial natural language understanding (NLU) research field which aims to accurately recognize reviewers' opinions on different aspects of products and services. Despite the prominence of recent ABSA applications, mainstream ABSA approaches inevitably rely on large-scale supervised corpora, and their final performances is susceptible to the quality of the training datasets. However, annotating sufficient data is labour intensive, which presents a significant barrier for generalizing a high-quality sentiment analysis model. Nonetheless, humans can make more accurate judgement based on their external background knowledge, such as factoid triples knowledge and event causality. Inspired by the investigations on external knowledge enhancement strategies in other popular NLP research, we propose a novel knowledge augmentation framework for ABSA, named the Oxford Dictionary descriptive knowledge-infused aspect-based sentiment analysis (DictABSA). Comprehensive experiments with many state-of-the-art approaches on several widely used benchmarks demonstrate that our proposed DictABSA significantly outperforms previous mainstream ABSA methods. For instance, compared with the baselines, our BERT-based knowledge infusion strategy achieves a substantial 6.42% and 5.26% absolute accuracy gain when adopting BERT-SPC on SemEval2014 and ABSA-DeBERTa on ACLShortData, respectively. Furthermore, to effectively make use of dictionary knowledge we devise several alternative knowledge infusion strategies. Extensive experiments using different knowledge infused strategies further demonstrate that the proposed knowledge infusion strategies effectively enhance the sentiment polarity identification capability. The Python implementation of our DictABSA is publicly available at https://github.com/albert-jin/DictionaryFused-E2E-ABSA.

1. Introduction

With the rapid development of computer science (CS) and artificial intelligent (AI) techniques , deep learning (DL) techniques (Jin et al., 2021), especially natural language processing (NLP), have yielded significant breakthroughs, such as relational extraction (Tao et al., 2019), event causality identification (Jian et al., 2020) and knowledge graph question answering (Jin et al., 2022). Of these, sentiment analysis tasks aim to identify people's attitudes, sentiments and appraisals within a piece of text (Abu Farha & Magdy, 2021; Zhou et al., 2019). A special branch of the sentiment analysis task, aspect-based sentiment analysis (ABSA) focuses

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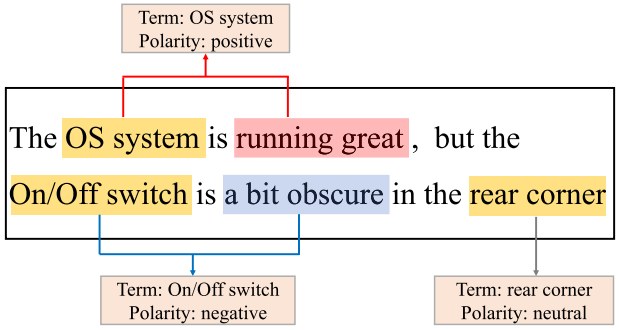


Fig. 1. A typical example for the Aspect-based Sentiment Analysis task.

on computing the sentiment polarity for each reviewer's opinion by considering the subjective terms and expressions which exist in explicit texts such as user reviews and social media posts (Zhang et al., 2018). As shown in Fig. 1, the laptop domain-related sentence "The OS system is running great but the On/Off switch is a bit obscure in the rear corner", is an example of a review in which the nouns "OS system" and "On/Off switch" are both aspect terms (or sentiment targets). Moreover, the reviewer expresses different sentiments in relation to these aspects, expressing a positive sentiment towards the "OS system" and a negative sentiment towards the "On/Off switch". Therefore, different to sentiment analysis which explores a general coarse-grained sentiment, ABSA is a more challenging task and there is a lack of large-scale training samples to ensure competitive performance.

Recently, the research focus on ABSA approaches has shifted to exploring a different model architecture, named "model augmentation" (Ju et al., 2021; Liang et al., 2021; Mukherjee et al., 2021) or providing large-scale annotated training data, named "data augmentation" (Hsu et al., 2021; Orbach et al., 2021). These researchers typically investigate new ABSA approaches on human-annotated examples only, which generally disregards leveraging extensive external knowledge for sentiment analysis reasoning. However, this resulted in the limitation that these methods generally lack external background knowledge (Xu et al., 2022). Trace back to human real life, the human are able to make appropriate judgements due to their long-term accumulated background knowledge. Also, humans can intuitively understand the sentence's semantics and sentiment better if the readers have acquired the sentence keywords' descriptive explanations.

The above-mentioned issues have motivated many researchers (Jian et al., 2020; Liao et al., 2022; Wang et al., 2022) to explore the emerging field of external knowledge fusion. These recent knowledge-infused improvements have been proven to be effective in achieving better intelligent model performance for some NLP tasks (Chen et al., 2022; Meškel & Frasincar, 2020; Zuo et al., 2020). These knowledge-infusion studies demonstrate the superiority of knowledge augmentation and the trend of exploring the different kinds of external knowledge infusion strategies.

Inspired by the superiority of the aforementioned external knowledge-injection method for natural language understanding, we investigate whether and to what extent infusing dictionary knowledge improves ABSA performance. We also propose a novel external knowledge augmentation strategy named the Oxford Dictionary descriptive knowledge infused aspect-based sentiment analysis (DictABSA), which can dynamically inject dictionary explanatory knowledge into a variety of ABSA architectures. Therefore, the external knowledge-enhanced approach DictABSA can better simulate the reasoning patterns of humans, and thus can derive an accurate semantic understanding and identify a precise sentiment polarity which is close to the ability of humans.

Specifically, our knowledge-infusion framework can be split conceptually into three main components, namely the *Dictionary Knowledge Retrieval* (DKR) module, the *Principal Original ABSA* (OriABSA) module and the *Auxiliary Knowledge Injecting* (KI) module:

(1) The Dictionary Knowledge Retrieval (DKR) module extracts the explanatory descriptions from the sentiment targets' dictionary from the official application programming interface[1] of the Oxford University Press.[2]

(2) After acquiring a large amount of external knowledge, the auxiliary knowledge injecting (KI) module comprises two mechanisms to incorporate knowledge into the token sequence generation and two mechanisms to incorporate knowledge into the feature encoder, which is in compliance with the hot-plugging criterion.

(3) Based on the infused dictionary knowledge encodings, multiple independent state-of-the-art ABSA models, which range from traditional pre-trained word embedding RNN-based approaches to the competitive pre-trained Transformer-based approaches, are utilized as the Original ABSA (OriABSA) component for the final ABSA prediction.

Fig. 2 schematically visualizes our approach. In the experiments, we adopt abundant model code implementations for each component and conduct a series of comparative analyses. In the following, we introduce the specific designing details of each component. We summarize this paper's contributions based on our experiments as follows:

1. Inspired by the significance of external knowledge-enhanced approaches, we utilize dictionary descriptive knowledge as a supervised signal and inject it into ABSA. To the best of our knowledge, we are the first to explicitly introduce external dictionary explanatory knowledge to the ABSA task.

[1] https://developer.oxforddictionaries.com/.

[2] https://languages.oup.com/research/oed-researcher-api/.

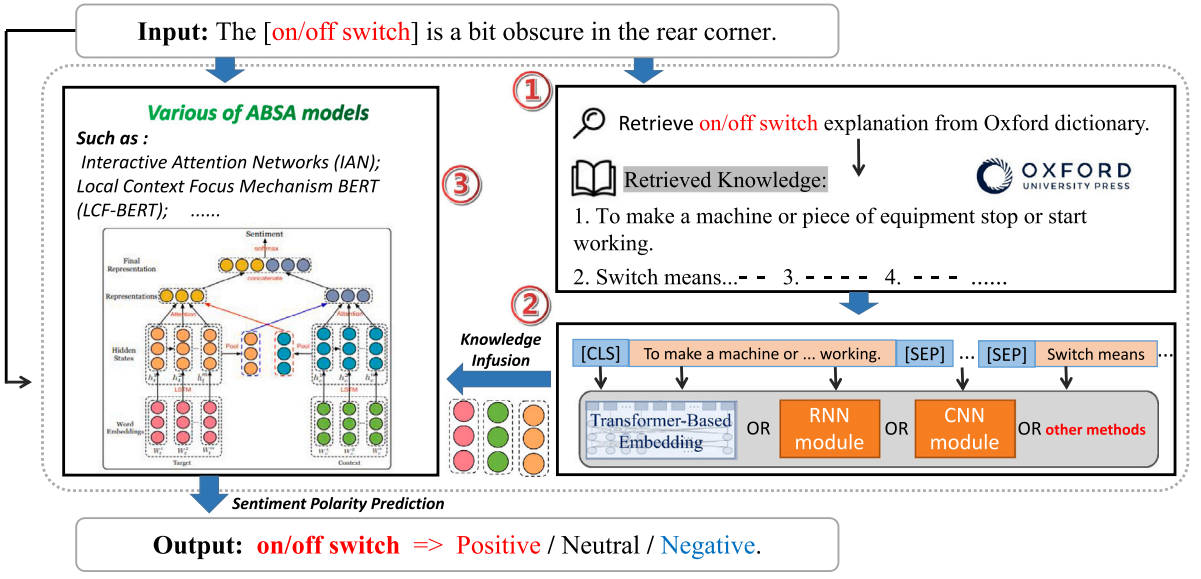


Fig. 2. The conceptual framework of our dictionary explanatory knowledge-enhanced DictABSA. Part. (1) denotes the *Dictionary Knowledge Retrieval Module*; Part. (2) denotes the auxiliary *Knowledge Injecting Module*; Part. (3) denotes the principal *Original ABSA Module*, respectively.

2. We devise a knowledge-enhanced joint ABSA framework which is called DictABSA. It concurrently incorporates the *Dictionary Knowledge Retrieval Module* (DKR) module, the *Knowledge Injecting Module* (KI) module and the *Original ABSA Module* (OriABSA) module.

3. To maintain the competitive efficiency and quick deployment of ABSA, we adopt the hot-plugging strategy into our framework. It improves the cohesion and decreases the coupling of the ABSA modules, which can simultaneously simplify and speed up the industry application deployment of ABSA.

4. We conduct extensive experiments to demonstrate that our DictABSA significantly outperforms previous ABSA approaches. We evaluate our knowledge infusion strategies by leveraging ten state-of-the-art ABSA methods on five benchmark datasets, proving the DictABSA's superiority over the previous ABSA methods that only relying on the supervised datasets.

The remainder of this paper is structured as follows: Section 2 introduces the previous literature on the ABSA task especially on the related approaches to external knowledge enhancement ABSA. Section 3 details the research preliminaries and objective of our proposed DictABSA. Section 4 describes the internal architecture of our proposed DictABSA and how it works in practice. Section 5 describes the experiments to quantitatively demonstrate the effectiveness of the designed knowledge-infusion framework. Next, we explicate how our dictionary knowledge-infusion strategies can handle some challenging cases with some case studies and error analyses provided in Section 6. Finally, we summarize our primary contributions and discuss the open challenges and possible promising advances in Section 8. The code (in Python 3, PyTorch, GoogleColab Jupyter notebook) is publicly available at GitHub: https://github.com/albert-jin/DictionaryFused-E2E-ABSA.

2. Related literature

Our work is closely related to aspect-based sentiment analysis (ABSA) and the knowledge-enhanced ABSA research. First, we briefly discuss the research about the ABSA works trained only on supervised corpora. Then, we introduce various external knowledge-enhancement studies on ABSA, which is the focus of our proposed DictABSA.

2.1. Aspect-based sentiment analysis

Artificial intelligent-based ABSA methods can be divided into traditional machine learning (ML)-based approaches (e.g. supported vector machine, Kiritchenko et al., 2014) and neural network (NN) deep learning-based approaches (e.g. CNN-based methods (Sun et al., 2019), RNN-based methods (Ma et al., 2017; Wang et al., 2016), PLM-based methods Ding et al., 2022; Emanuel & Marcacini, 2021). However, ML-based approaches are based on heavy feature engineering (Fry et al., 2020; Min et al., 2021; Zhou et al., 2019), which wastes much time analysing source corpora and manual-craft features. Compared with NN-based methods, ML-based methods are more burdensome, inefficient and lack domain transferability.

Hochreiter and Schmidhuber (1997) firstly employ long short-term memory (LSTM) to sentiment analysis. Tang, Qin, Feng et al. (2016) propose TC-LSTM and TD-LSTM, in which several simple but efficient strategies, such as concatenation, are adopted to consider the interaction between the aspect term and its sentiment context. Wang et al. (2016) propose attention-based LSTM

with an aspect embedding-based long short-term memory network (ATAE-LSTM). Tang, Qin, Liu (2016) introduce a deep memory network (MemNet), which utilizes multi-hop computational memory units to handle the aspect term-based sentiment classification. Ma et al. (2017) introduce interactive attention networks (IAN) to interactively learn the attention between the targets and contexts by leveraging a novel interactive attention mechanism. Liu et al. (2018) propose content attention-based aspect-based sentiment classification (CABASC), which comprises two special attention mechanisms for both global and local attention. BERT-sentence pair classification (BERT-SPC) (Devlin et al., 2019) transforms the ABSA task into a sentence pair classification problem on top of the Bidirectional Encoder Representations from Transformers (BERT). Inspired by the disentangled representation learning (He et al., 2020; Locatello et al., 2019) investigate the effectiveness of the Disentangled Representation Learning BERT for ABSA by substituting the feature encoder BERT with Decoding-enhanced BERT with Disentangled Attention (DeBERTa). To alleviate the problem of training data deficiencies, Hsu et al. (2021) propose a data augmentation approach for ASBA by replacing the unimportant words while preserving aspect-level polarity. Xie et al. (2020) present a semi-supervised ABSA approach based on advanced data augmentation strategies such as RandAugment and back-translation.

2.2. External knowledge-enhanced ABSA

In addition to ABSA approaches only trained on supervised corpora, external knowledge-enhanced ABSA methods have been extensively explored recently. Various types of external knowledge have been studied to incorporate into ABSA, such as sentiment lexicons (Fan et al., 2019; Teng et al., 2016; Zou et al., 2018), syntactic dependency trees (Sun et al., 2019), and knowledge graphs (Ghosal et al., 2020; Zhang et al., 2020).

Tang et al. (2014) firstly incorporate the external knowledge, sentiment-specific word embedding (SSWE), to Twitter sentiment classification by integrating the sentiment information into the loss functions. Sentiment-specific word embedding is learned from large-scale distant-supervised tweets (about 10 million) selected by positive and negative emotions. Teng et al. (2016) investigate the effectiveness of introducing sentiment lexicon features to state-of-the-art recurrent neural network (RNN) models for sentiment analysis. They propose a context-sensitive lexicon-based model based on a simple weighted-sum mechanism. Furthermore, the model improves the cross-domain robustness of RNN for sentiment analysis. Lei et al. (2018) propose a multi-sentiment resource-enhanced attention network (MEAN) to alleviate the problem by integrating three kinds of sentiment linguistic knowledge (e.g., sentiment lexicon, negation words, intensity words) into the deep neural network via attention mechanisms. Zou et al. (2018) make use of sentiment lexicon knowledge and thus propose a novel lexicon-based supervised attention model (LBSA). Based on the linguistic sentiment lexicons, it can simultaneously focus on the sentiment content and generate sentiment-informative representations. Fan et al. (2019) propose a regularized hierarchical neural network (RHNN) which incorporates both sentiment lexicon knowledge and common knowledge. RHNN aggregates discourse context information through a hierarchical learning structure and restrains the parameters with knowledge-based regularizations. Bao et al. (2019) propose a flexible and robust knowledge-enhanced model based on an attention LSTM neural network, which leverages numerical polarity features provided by existing lexicon resources. Margatina et al. (2019) introduce three novel attention-based conditioning methods which incorporate external knowledge to augment the performance of sentiment analysis, emotion recognition, irony, and sarcasm detection. Specifically, the external knowledge they leveraged includes lexica psycho-linguistic, sentiment and emotion annotations. Ke et al. (2020) propose a novel language representation model, called SentiLARE, which introduces word-level linguistic knowledge from SentiWordNet into PLMs, including context-aware sentiment polarity and part-of-speech tag for each word. Tian et al. (2020) propose sentiment knowledge enhanced pre-training (SKEP), a sentiment knowledge enhanced pre-training sentiment analysis method which integrates different types of sentiment knowledge together, such as sentiment words and aspect-sentiment pairs.

Sun et al. (2019) explore the possibility of integrating a syntactic dependency tree to aspect-level sentiment analysis. They exploit bi-directional long short-term memory (Bi-LSTM) to learn sentence embeddings and further enhance sentence representations by the sentence's dependency tree which is processed by a graph convolutional network (GCN). Zhang et al. (2020) propose a semantic-emotion knowledge transferring (SEKT) model which uses external knowledge (semantic and emotion lexicons) as a bridge to enable knowledge transfer across different targets. SEKT constructs a semantic-emotion heterogeneous graph from external semantic and emotion lexicons and then feeds this into a graph convolutional network to learn the multi-hop semantic connections between words and emotion tags. Ghosal et al. (2020) devise an external commonsense knowledge framework called KinGDOM which adapts an external knowledge graph to the cross-domain sentiment analysis task. KinGDOM utilizes ConceptNet to enrich the semantics of a document by providing both domain-specific and domain-general background concepts.

3. Preliminaries

Before describing our specific external dictionary descriptive knowledge-infusion solution, we formulate the mainstream ABSA modelling paradigms in the following sections. The research objective of this work aims at predicting the sentiment polarity for a specific aspect term within a language expression, which corresponds to the fourth definition, aspect sentiment classification (ASC) (Zhang et al., 2021; Zhou et al., 2019).

3.1. Single ABSA tasks

Aspect Term Extraction (ATE): Aspect term extraction aims to extract the aspect expressions with which users express opinions for a given sentence (Zhang et al., 2018). Given an N-length input sentence $S = \{w_i\}_{i=1}^{N}$, ATE aims to detect all aspect terms' beginning and ending positions $T_{ate} = \{[P_i^{start}; P_i^{end}]\}_{i=1}^{m}$, where P_i^{start} and P_i^{end} are the beginning position and the ending position of the mth aspect term, respectively; m is the count of all aspect terms. The ATE task is always categorized into a sequence tagging process by utilizing a set of pre-defined labels $\{B, I, O, S, E\}$, in which B, I, O, S, E denote the aspect terms' beginning/inside/other/single-word/ending boundary position. For example: in the sentence "*I'm overall pleased with the interface and the portability of this product.*", ATE can extract three aspect terms: "*interface*", "*portability*" and "*product*".

Aspect Category Detection (ACD): Aspect category detection aims at recognizing the discussed categories of each aspect term for a given sentence, where the categories belong to a pre-defined category set and is often domain-specific. Given a pre-defined set $\{c_{i=1}^{M}\}$ which contains M aspect categories, a sentence $S = \{w_i\}_{i=1}^{N}$ with N words and its aspect terms $\{a_{i=1}^{k}\}$, ACD infers all the aspect terms' belonged categories $\{e_{i=1}^{k}\}$ for each aspect term from $\{a_{i=1}^{k}\}$, where the $e_i \in \{c_{i=1}^{M}\}$. For example, given a sentence "*I had to install Mountain Lion and it took a good two hours.*", ACD is expected to cluster "*Mountain Lion*" to the aspect category "*Electronics*".

Opinion Term Extraction (OTE): Opinion term extraction is similar to aspect term extraction, which aims to identify the boundaries of all sentiment-related expressions towards the aspect terms since the opinion expressions always co-occur with the aspect terms. The task formulation is similar to the above aspect term extraction task and can also be formulated as a sequence labelling task. For the example "*Shipped very quickly and safely.*", OTE attempts to extract all the opinion terms "*quickly*" and "*safely*".

Aspect Sentiment Classification (ASC): Aspect sentiment classification aims to classify the specific aspect's sentiment polarity within a sentence from POS/NEU/NEG sentiment polarities. Formally, Given an N-length input sentence $S = \{w_i\}_{i=1}^{N}$ and the kth aspect term $\{w_j\}_{j=m}^{n}$ in this sentence where m denote the start position of the kth aspect term and n denote the end position of the kth aspect term, ASC aims to inference a tag type y_i^{asc}, where $y_i^{asc} \in \{pos, neg, neu\}$. For example, given a sentence "*The MBP is beautiful has many wonderful capabilities.*" and its exploited aspect terms "*MBP*" and "*capabilities*", ASC is expected to predict sentiment type [POSITIVE] for both sentiment element "*MBP*" and "*capabilities*".

3.2. Compound ABSA tasks

In addition to the aforementioned single ABSA tasks, there are also several mainstream compound ABSA tasks: (1) aspect-opinion pair extraction which combines sub-task aspect term extraction with sub-task opinion term extraction; (2) end-to-end ABSA which combines sub-task aspect sentiment classification with sub-task Aspect Term Extraction; (3) aspect category sentiment analysis which combines sub-task aspect category detection with sub-task aspect term extraction; (4) aspect sentiment triplet extraction which aims to identify the opinion target; what its sentiment orientation is; and why such a sentiment is expressed; (5) aspect sentiment quad prediction which aims at identifying all the four sentiment elements in quadruplet form (e.g. {food, sandwich, POSITIVE, delicious}) given a text item; (6) aspect-category sentiment detection which aims at detecting the target categories and the sentiment polarities of all aspect terms for a given sentence (Zhang et al., 2018).

4. Our proposed model

In light of our framework structure and the above preliminaries, we describe each module's working mechanism in detail: Section 4.2 introduces the *Dictionary Knowledge Retrieval* (DKR) module; Section 4.3 introduces the *Principal Original ABSA* (OriABSA) module; Section 4.4 introduces the *Auxiliary Knowledge Injecting* (KI) module.

4.1. Architecture overview

To decrease the computation cost of artificial intelligence applications and to easily apply them in practice, the hot-plugging strategy is a popular and important model architecture designing concept. It focuses on improving the cohesion degree inside each AI application's module while decreasing the coupling degree between the AI application's modules. No matter which module structure is adopted, other modules can still retain their original features and functions.

To investigate which one of the configurations is the most suitable and the impact of different specific realizations, we substitute ten different implementations as the *Principal Original ABSA* module and four different actual strategies as the *Auxiliary Knowledge Injecting* module in our experiments. Our dictionary descriptive knowledge infusion framework and the architecture of its major components are illustrated in Fig. 2. The details are as follows.

Algorithm 1 The Oxford Dictionary Knowledge Retrieval Steps

Require: Request the Oxford Dictionary official API /*entries*/{*language*}/{*aspect_term*} within request headers which contains api_id and api_key.

Ensure: The return dictionary entries which includes many related commonsense knowledge in **JSON** format.

1: {id: sandwich, metadata: {operation: retrieve, provider: Oxford University Press, schema: RetrieveEntry}, results: [{id: sandwich, language: en-gb, lexicalEntries: [{entries: [{etymologies: [mid 18th century: named after the 4th Earl of Sandwich (1718–92), an English nobleman said to have eaten food in this form so as not to leave the gaming table], pronunciations: [{audioFile: ...mp3, dialects: [British English], phoneticNotation:..., phoneticSpelling:...}, ...], senses: [{definitions: [an item of food consisting of two pieces of bread with a filling between them, eaten as a light meal], domainClasses: [{id: savoury, text: Savoury}], examples:[{text: a ham sandwich}], id: m_en_gbus0898220.007, semanticClasses: [...], shortDefinitions: [insert or squeeze (someone or something) between two other people or things, ...], subsenses: ..., }, {definitions: [something that is constructed like or has the form of a sandwich.], ...}], synonyms: [{language: en, text: light meal}, ...

LAST: As shown in the above red colored part, extract the glossaries and definitions.

Table 1
Demonstration of the original dataset before dictionary descriptive knowledge retrieval.

Sentence	Aspect term	Sentiment polarity
The lobster sandwich is good and the spaghetti with scallops and shrimp is great.	sandwich	1
The lobster sandwich is good and the spaghetti with scallops and shrimp is great.	spaghetti	1

Table 2
Demonstration of the descriptive knowledge-enhanced dataset.

Sentence	Aspect term	dictionary_knowledge	Sentiment polarity
The lobster sandwich is ...	sandwich	item of food consisting ... [SEP] insert or squeeze ...	1
... and the spaghetti ... is ...	spaghetti	an Italian dish ... [SEP] pasta made in solid ...	1

4.2. Dictionary knowledge retrieval

In Computer Vision (CV) (He et al., 2020; Jin et al., 2021), image feature translation such as colour/contrast changes and object crop/rotation can be regarded as a unique explanatory factor to CV tasks. This explanatory knowledge not only leads to better performance, it can also can make the generated representations more human-interpretable. To the best of our best knowledge, this is the first attempt to bring this concept into the ABSA task to increase data diversity and exploit the effect of explanatory knowledge (dictionary knowledge) infusion.

Firstly, given an instance example of ABSA and its extracted aspect term set $\{w_{i=1}^{N}\}$, we request the Oxford Dictionary API[3] of the Oxford University Press[4] to obtain all explanatory definitions for each aspect term w_i. More specifically, the Oxford API provides many dictionary-related sub-routers, in which the sub-router '/api/v2/entries/{language}/{aspect_term}'[5] is what we actually need. Taking the aspect term "*sandwich*" as an example, we retrieve the term's meanings and extract the necessary definition-related knowledge from the API response. The related procedure is shown in Algorithm 1.

Secondly, after obtaining each aspect term's descriptive knowledge of the given sentence, we merge all definitions of an aspect term into one expression string by inserting "[SEP]" between these definitions. Then, we prepare the knowledge-enhanced dataset, an original training instance $\{sentence : ..., aspectterm : ..., sentimentpolarity : ...\}$ would be converted to a new data format $\{sentence : ..., aspectterm : ..., dictionary_knowledge : ..., sentimentpolarity : ...\}$. Take the sentence "The lobster [sandwich] is good, and the [spaghetti] with scallops and shrimp is great" as an example, Fig. 3 illustrates the detailed data enhancing procedure based on the original data shown in Table 1. Meanwhile, Table 1 gives a brief instance example for this original dataset, and Table 2 is the expanded dataset example after retrieving the Oxford Dictionary API.

4.3. Principal original ABSA

Based on the obtained descriptive knowledge, we introduce several mainstream state-of-the-art ABSA approaches. As shown in Fig. 2, the corresponding *Principal Original ABSA* module is designed to identify the sentiment polarity of the aspect terms by the incorporated knowledge feature generated from the Auxiliary Knowledge Injection module (the workings of this module are

[3] https://developer.oxforddictionaries.com/.

[4] The dictionary knowledge we downloaded throughout the experiments has allowed to us through official formal channels. Because we have joined the *Unlimited Plan* which is offered by Oxford University Press and have gotten the service license of Oxford Dictionary. The *Unlimited Plan* provides free access to the Oxford Dictionaries API.

[5] Making a GET request to https://od-api.oxforddictionaries.com/api/v2/entries/en-gb/sandwich in POSTMAN.

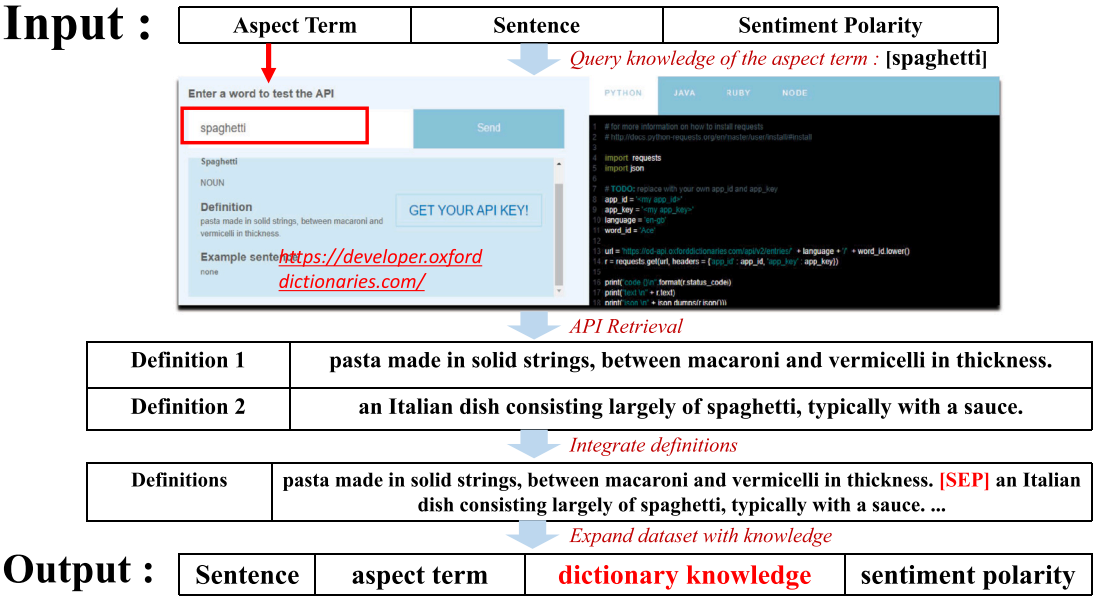


Fig. 3. The data flow diagram of the dictionary knowledge retrieval module (take "*spaghetti*" as the aspect term).

explained in the next part), which demonstrates that this module plays an important role in the whole ABSA task. The major characteristic of our DictABSA is the adopted hot-plugging strategy, which can effectively decrease the coupling degree between the *Principal Original ABSA* module and the *Auxiliary Knowledge Injection* module. Therefore, the different knowledge infusion mechanisms will not be in conflict with any of the adopted original ABSA models.

Specifically, in our experiment, we adopt ten different ABSA approaches to demonstrate that no matter which ABSA approaches is used, infusing dictionary knowledge is helpful in general. These ten models can be divided into three categories: RNN-based models (Hochreiter & Schmidhuber, 1997; Tang, Qin, Feng et al., 2016; Wang et al., 2016), pre-trained Transformer BERT-based models (Devlin et al., 2019; Zeng et al., 2019) and ABSA-DeBERTa[6] (Emanuel & Marcacini, 2021) (a BERT variant which uses disentangled attention mechanism He et al., 2020).

We introduce the *Principal Original ABSA* module in detail by taking several specific experimental implementations as examples. To further analyse the principle of how the hot-plugging strategy works and how the ABSA model utilizes external knowledge from the *Auxiliary Knowledge Injection* module, we select three a representative method from each of these three categories as follows.

4.3.1. Interactive attention networks (IAN)

Our adopted RNN-based models are conventional neural networks which couple the pre-trained fixed token embeddings and regard RNNs, such as LSTM, Bi-LSTM and GRU, as the ABSA backbone (e.g. GloVe Pennington et al., 2014 and Word2Vec Mikolov et al., 2013), with a well-designed task-specific neural architecture. In our comparison experiments, we use seven RNN-based models as the *Principal Original ABSA* module. The five RNN-based ABSA methods are primitive LSTM (Hochreiter & Schmidhuber, 1997) (bare LSTM without any other layers), ATAE-LSTM (Wang et al., 2016) (a special attention strategy is introduced to pay more attention to the relative context of the targeted aspects), IAN (the representative RNN-based model which we will analyse) (Ma et al., 2017), TC-LSTM and TD-LSTM proposed by Tang, Qin, Feng et al. (2016) (they divide the inputted contexts into the left part and the right part by splitting the aspect term in a sentence), MemNet (Tang, Qin, Liu, 2016) (multi-hop attention memory-based deep memory network), and CABASC (Liu et al., 2018) (aspect-based sentiment classification model based on content attention).

Only considering the sentence context solely will inevitably miss the semantics of the aspect terms which are relatively lacking in semantics. As shown on the left of Fig. 4, IAN first takes the aspect term and its sentence context as inputs and generates the hidden states of the aspect term with the hidden states of its context (the whole sentence) by two LSTM networks independently. After the textual encoding process, IAN learns the interactive feature representation of the hidden states of the targeted aspect and its corresponding contexts by adopting a new designed interactive attention mechanism. The interactive attention mechanism extracts the important interactive feature of the aspect term and its context. Specifically, the interactive attention mechanism firstly computes the average weight of the aspect term representation and context representation. Then, IAN uses the average weight to supervise the generation of the attention vectors. The attention mechanism then captures the important features for both the aspect term and the context. Finally, the aspect term and the context can influence each other so that IAN can generate a more comprehensive feature representation by this IAN design. Lastly, the aspect term and context representation are merged as the final ABSA feature

[6] https://github.com/huberemanuel/DeBERTa/.

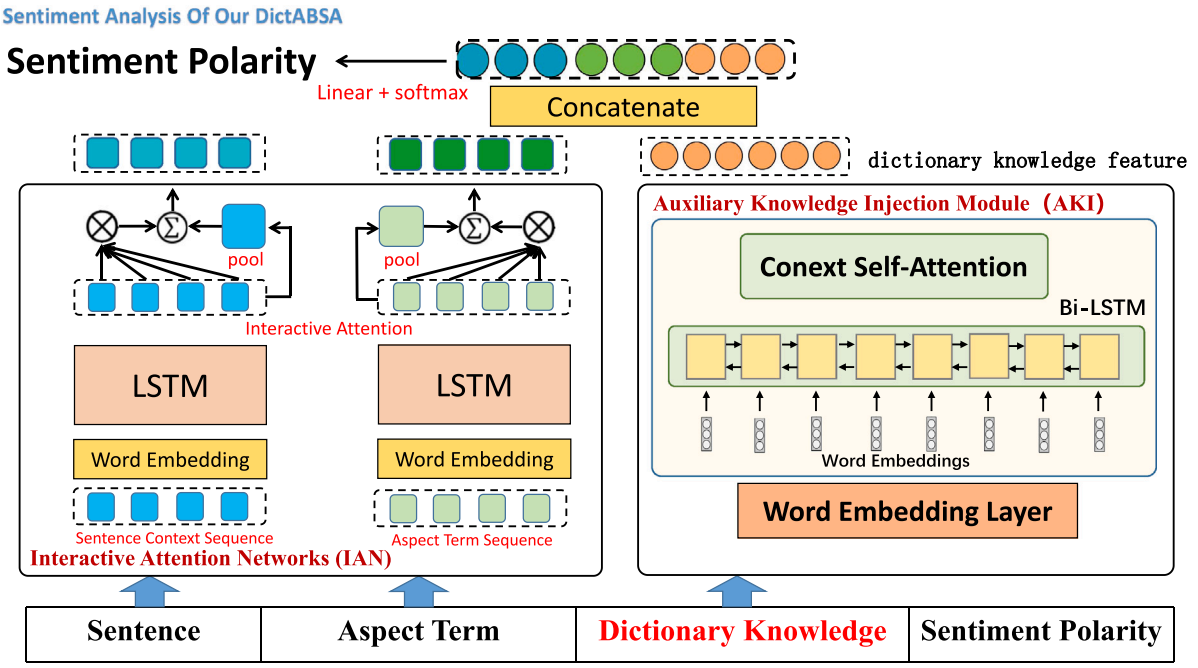


Fig. 4. The architecture visualization of IAN with an RNN-based external knowledge infusion mechanism. The left side denotes the Principal Original ABSA module; the right side denotes the Auxiliary Knowledge Injection module (using Bi-LSTM & attention encoder); the IAN model structure is copied from the original IAN paper.

representation by the vector concatenation operation. Then, the last feature representation is fed to a *softmax* operation to evaluate the probability of the sentiment polarity being 'neutral', 'positive' or 'negative'.

The above introduction relates to the original ABSA inference procedure (refer to paper[7] for more details).

As shown on the right of Fig. 4, the *Auxiliary Knowledge Injection* module takes the Bi-LSTM and attention mechanism as an example. Here, we pay attention to the module's external interface rather than the internal details of the *Auxiliary Knowledge Injection* because the internal construction details are not the focus of this section. The alternative *Auxiliary Knowledge Injection* module is analysed in Section 4.4.

The objective of the *Auxiliary Knowledge Injection* module is to provide the OriABSA module with the incorporated dictionary descriptive knowledge. Firstly, the *Auxiliary Knowledge Injection* module takes the dictionary knowledge column of the ABSA dataset as input. Each token in the knowledge description is converted into a feature embedding through the word embedding layer. Secondly, the Bi-LSTM layer learns the hidden representation and then the *Self-Attention Unit* is utilized to calculate the global representation as it is more effective and robust in the semantic extraction and aggregation of the natural language text. Lastly, the encoded representation (in the top right of Fig. 4), which denotes the captured knowledge semantic, is concatenated with the original ABSA representation (in the top left of Fig. 4) for the final sentiment analysis.

Overall, the *Auxiliary Knowledge Injection* module can be regarded as a black box which is responsible for knowledge incorporation and provides the *Principal Original ABSA* with incorporated external knowledge. In other words, it is easy and efficient to switch to alternative solutions for the *Auxiliary Knowledge Injection* module.

4.3.2. Local context-focused BERT for ABSA (LCF-BERT-ABSA)

In addition to RNN-based models which gradually reach a bottleneck, complicated pre-trained language models (PLMs), such as BERT, RoBERTa and XLNet have brought considerable advancements to a wide range of natural language processing tasks (Devlin et al., 2019). Motivated by the observation that PLM-based models are substantially superior in tackling the sentence pair classification task which is similar to sentiment analysis tasks, we also attempt to introduce the Transformer BERT-based model as the replacement of the *Principal Original ABSA* module. One alternative is BERT for sentence pair classification (BERT-SPC),[8] which converts the sentiment analysis task to the sentence pair classification problem by regarding the inputs as two segments by appending the aspect term to the context. The other alternative is the aspect-based sentiment classification model by using local

[7] https://arxiv.org/pdf/1709.00893.
[8] https://arxiv.org/pdf/1810.04805.pdf.

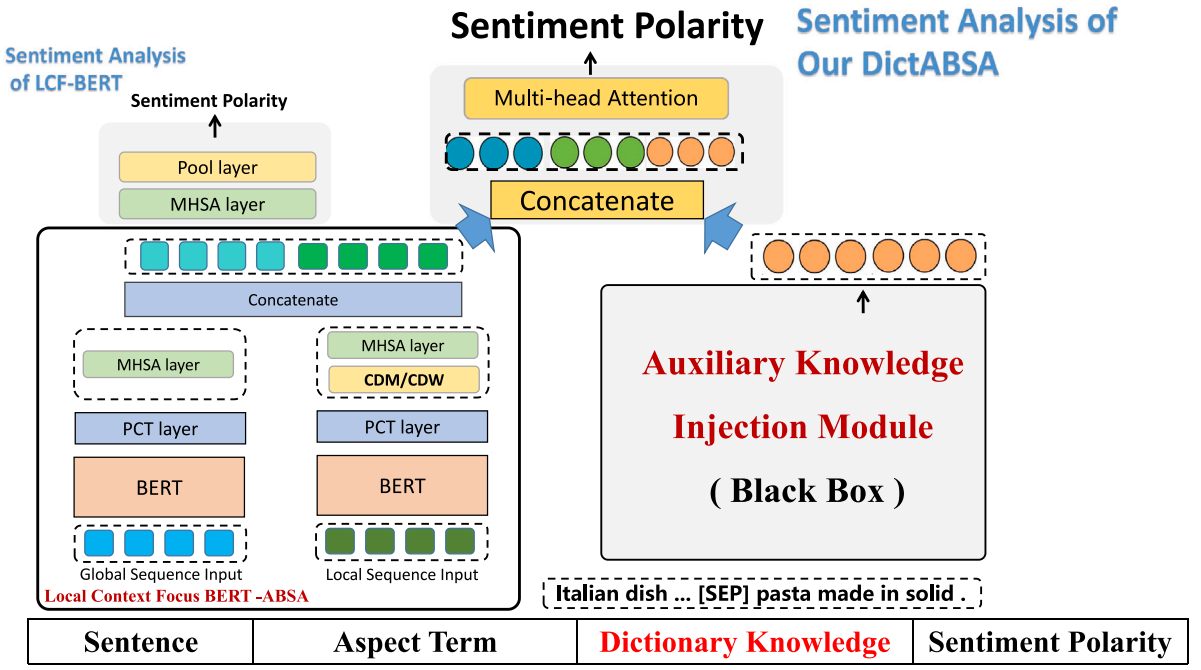


Fig. 5. The implementation of the *Principal Original ABSA* module by adopting LCF-BERT-ABSA. The auxiliary KI module can be regarded as a black box here.

context-focused BERT (LCF-BERT),[9] which uses the original BERT fine-tuning strategy and adds a local context to emphasize the relative distance between tokens.

Having considered the superiority of LCF-BERT-ABSA, we first choose the representative model LCF-BERT-ABSA as an explanatory example to further investigate the reference activities of the PLM-based model.

As shown on the left of Fig. 5, LCF-BERT leverages two different BERT-encoders to respectively model the aspect term's local features and the sentence's global features. The BERT-encoder is powerful enough to recognize the context semantics (Devlin et al., 2019). For all inputs of the local target representation I^{local} and the global context representation I^{global}, the corresponding output representations O^{local} and O^{global} are calculated as follows:

$$O^{local}_{BERT} = BERT^{l}\left(I^{local}\right) \quad \& \quad O^{global}_{BERT} = BERT^{g}\left(I^{global}\right) \tag{1}$$

O^{local}_{BERT} and O^{global}_{BERT} are the encoded feature representations from the BERT extractors. $BERT^{l}$ and $BERT^{g}$ are the corresponding BERT extractors for the local aspect term and global sentence context, respectively.

Then, as illustrated in Fig. 5, the position-wise convolution transformation (PCT) and multi-head self-attention (MHSA) is appended to the local context focus layer in the following:

$$O^{local}_{\text{mhsa}} = MHSA^{l}\left(O^{local}_{\text{bert}}\right) \quad \& \quad O^{g}_{PFE} = PCT^{l}\left(O^{local}_{\text{mhsa}}\right) \tag{2}$$

$$O^{global}_{\text{mhsa}} = MHSA^{g}\left(O^{global}_{\text{bert}}\right) \quad \& \quad O^{g}_{PFE} = PCT^{g}\left(O^{global}_{\text{mhsa}}\right) \tag{3}$$

where the variables O^{global}_{bert} and O^{global}_{bert} denote the last BERT layer's hidden states; PCT^{l} and PCT^{g}, $MHSA^{l}$ and $MHSA^{g}$ denote the position-wise convolution transformation operation, and the multi-head self-attention operation for the local context and global context, respectively. O^{l}_{PFE} and O^{g}_{PFE} denote the output feature vectors of the local context and the global context generated by the pre-feature extractor layer.

Finally, the local context-focused (LCF) mechanism adopts a feature extractor network which is equipped with a LCF layer and an MHSA encoder to learn the local and global contexts' semantic features. The LCF layer will mask or weight down the output position (POS) features above the semantic-relative distance (SRD) threshold to which the dotted arrow points. Meanwhile, It completely preserves the output position that the solid arrow points to. Due to the applied LCF strategy, LCF-BERT outperforms most SOTA methods in many well-known ABSA benchmarks (Pennington et al., 2014; Zeng et al., 2019).

Similar to the configuration of adopting IAN as the *Principal Original ABSA*, which is shown in Fig. 4, the hot-plugging strategy allows us to regard the *Auxiliary Knowledge Injection* module as a black box by only utilizing the knowledge input interface and

[9] https://github.com/yangheng95/LCF-ABSA.

the vector representation output interface of the *Auxiliary Knowledge Injection*. Also, the dictionary descriptive knowledge is fed to the KI module, and the KI generates the hidden representation of the knowledge. Then, we apply a tensor concatenation for the representations generated from the *Principal Original ABSA* and the *Auxiliary Knowledge Injection* module. Following BERT's original fine-tuning and downstream classification strategies, we adopt a full-connection network with a standard *softmax* operation to identify sentiment polarity. The classification loss calculated by the cross-entropy function is used to adjust the initial parameters of the overall framework.

4.3.3. ABSA-DeBERTa

Due to the language knowledge obtained in the pre-training procedure, the PLM-based fine-tuning strategy significantly outperforms the previous methods for the ABSA task. In addition to the Transformer BERT-based approaches, recent pre-trained Transformers like RoBERTa, XLNet, and DeBERTa (He et al., 2020) also brought substantial performance gains to the ABSA task. For instance, as shown in Fig. 6, Emanuel and Marcacini (2021) investigate the use of disentangled representation learning (Locatello et al., 2019) for the ABSA task. The disentangled attention introduced by He et al. (2020) separates the content and text position components and learns the attention weights for each component. For a ith token T_i in a sequence, the disentangled attention mechanism represents it using two vectors: *content-focus* $\{H_i\}$ and *relative position-focus* $\{P_{i|j}\}$, where $\{P_{i|j}\}$ is the relative-position vector representation of token i and j. After the discrete representation, DeBERTa applies a decomposition operation over each token pair by a cross-attention (disentangled) matrix $M_{i,j}$ in Eq. (4).

$$\begin{aligned} M_{i,j} &= \left\{\boldsymbol{P}_{i|j}, \boldsymbol{H}_i\right\} \times \left\{\boldsymbol{P}_{j|i}, \boldsymbol{H}_j\right\}^{\top} \\ &= \boldsymbol{P}_{i|j}\boldsymbol{P}_{j|i}^{\top} + \boldsymbol{P}_{i|j}\boldsymbol{H}_j^{\top} + \boldsymbol{H}_i\boldsymbol{P}_{j|i}^{\top} + \boldsymbol{H}_i\boldsymbol{H}_j^{\top} \end{aligned} \tag{4}$$

where the $\boldsymbol{H}_i\boldsymbol{H}_j^{\top}$, $\boldsymbol{H}_i\boldsymbol{P}_{j|i}^{\top}$, $\boldsymbol{P}_{i|j}\boldsymbol{H}_j^{\top}$, and $\boldsymbol{P}_{i|j}\boldsymbol{P}_{j|i}^{\top}$ denote the *content interactive*, the *content-with-position*, the *position-with-content*, and the *position interactive* attention scores, respectively.

In addition, to accurately take the absolute position of the word into consideration in natural language sequence modelling, an additional token-masking decoding layer is introduced to inject the absolute position features in the pre-training model decoding procedure. Moreover, to improve the models' generalization, a new virtual adversarial learning scheme is used to fine-tune Transformer-based encoders to the NLP downstream sub-tasks.

These novel techniques increase the neural network pre-training efficiency significantly and improve the performance on the downstream tasks of both natural language understanding (NLU) and natural language generation (NLG). Following them, we migrate DeBERTa to substitute the module *Principal Original ABSA*, which is shown in the left part of Fig. 6. For an input sequence for which the sentiment is needed to predict, after feeding the sentence and the aspect term into the DeBERTa encoder, we can obtain the context's and target's initial representations by the feedforward DeBERTa encoding procedure. Also, we regard the *Auxiliary Knowledge Injection* module as a black box and leverage its generalized vectors for the dictionary descriptive knowledge representation.

Then, we concatenate the pre-trained language model DeBERTa's output vector with the output vector of the *Auxiliary Knowledge Injection* module. The final concatenated representation is denoted as $H_{\{content;knowledge\}}$ and is fed to the downstream sentiment classification task since the $H_{\{content;knowledge\}}$ naturally captures both the context semantics and the noun-explanatory commonsense knowledge.

To make the final sentiment analysis, we follow the traditional sentiment polarity classification approaches. We adopt a full-connect layer to obtain the logits and perform the sentiment category prediction by a standard *softmax* layer and adopt the loss function: categorical cross-entropy loss. The corresponding formulas are as follows:

$$\mathbf{L}_{\{c;k\}}^{(i)} = \sigma\left(\mathbf{W}\mathbf{H}_{\{content;knowledge\}}^{(i)} + \mathbf{b}\right) \tag{5}$$

$$\mathbf{P}^{(i)} = \mathrm{SoftMax}\left(\mathbf{L}_{\{c;k\}}^{(i)}\right) \quad where \quad \mathbf{P}^{(i)} = [\mathbf{p}_{\{1\}}^{(i)}; \mathbf{p}_{\{2\}}^{(i)}; ...\mathbf{p}_{\{M\}}^{(i)}] \tag{6}$$

$$\mathbf{Loss}^{CE} = \frac{1}{N}\sum_{i=1}^{N}\mathrm{Loss}^{(i)} = -\frac{1}{N}\sum_{i=1}^{N}\left(\sum_{k=1}^{M} GL_k^{(i)} * \log\left(\mathbf{p}_k^{(i)}\right)\right) \tag{7}$$

where the variables $\mathbf{W}$ and $\mathbf{b}$ represent the parameters of the full-connect layer; σ is the $SigMoid$ activate function; $\mathbf{L}_{\{c;k\}}^{(i)}$ denotes the output logits; $\mathbf{P}^{(i)}$ is the three-dimension probability matrix which conspires the predicted probability value ($0 < \mathbf{p} < 1$ and $k \in \{1, 2, \ldots \mathbf{M}\}$) of belonging to POSITIVE/NEUTRAL/NEGATIVE sentiment polarity, $\mathbf{p}_{\{1\}}^{(i)}$, $\mathbf{p}_{\{2\}}^{(i)}$, and $\mathbf{p}_{\{3\}}^{(i)}$, in which $\mathbf{M}$ is the count of the sentiment polarity category which is set to 3 as the default; $GL_k^{(i)}$ denotes the ground truth sentiment polarity for the ith training sample in the dataset; $\mathbf{N}$ is the dataset's sample number. $\mathbf{Loss}^{CE}$ denotes the cross-entropy loss function with $L2$ regularization operation. Lastly, we adopt the loss back-propagation algorithm to calculate the gradients and optimize the network's model parameters. The sentiment polarity category within the highest predicted probability is regarded as the final sentiment polarity.

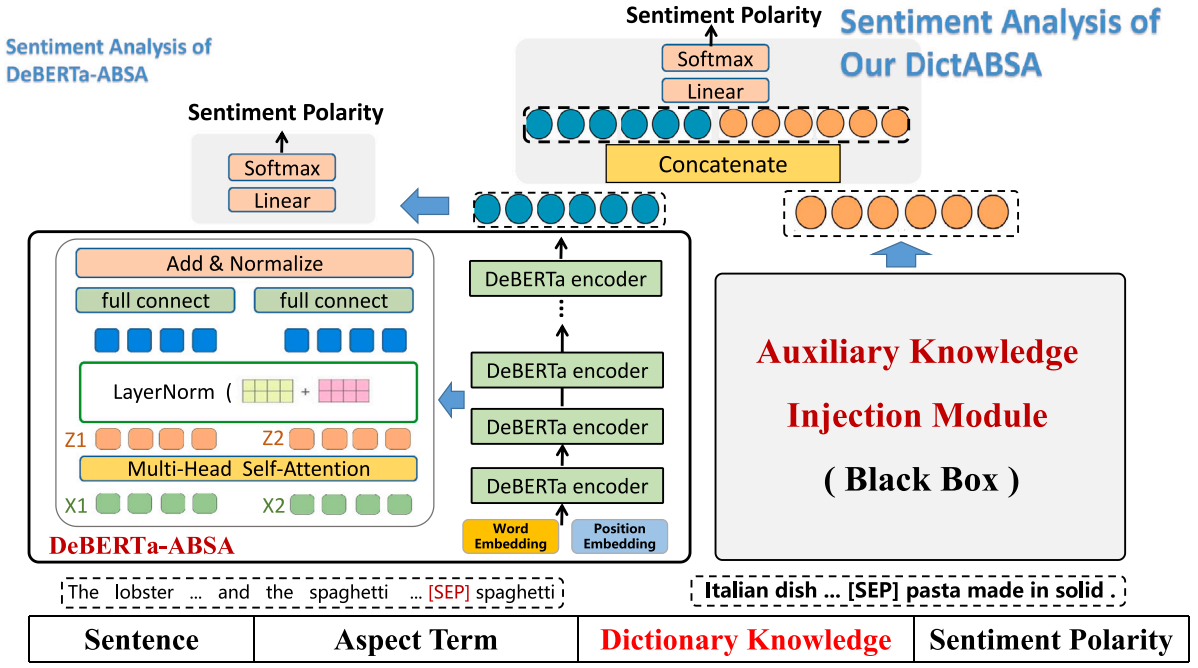


Fig. 6. The specific implementation of the Principal Original ABSA module by adopting ABSA-DeBERTa. The auxiliary module KI can be regarded as a black box here.

4.4. *Auxiliary knowledge injection*

Due to the adopted hot-plugging strategy, the *Auxiliary Knowledge Injection* module can be substituted by a variety of knowledge incorporation approaches. To better investigate the effect of dictionary knowledge on the ABSA task, we implement several different knowledge infusion strategies, as shown in Fig. 7, including two methods of incorporating the knowledge into the token sequence (e.g. concatenate, insert) and two methods of incorporating the knowledge into the feature encoder (e.g. RNNs, BERT).

Fig. 7(a) illustrates the specific details of our proposed two token sequence incorporation methods: one way is the token sequence concatenation (bottom of Fig. 7(a)), where we directly concatenate the natural language expressed dictionary knowledge and the original sentence; the other way is the token sequence insertion (top of Fig. 7(a)), where we directly insert the natural language expressed dictionary knowledge into the original position in the sentence of the aspect term.

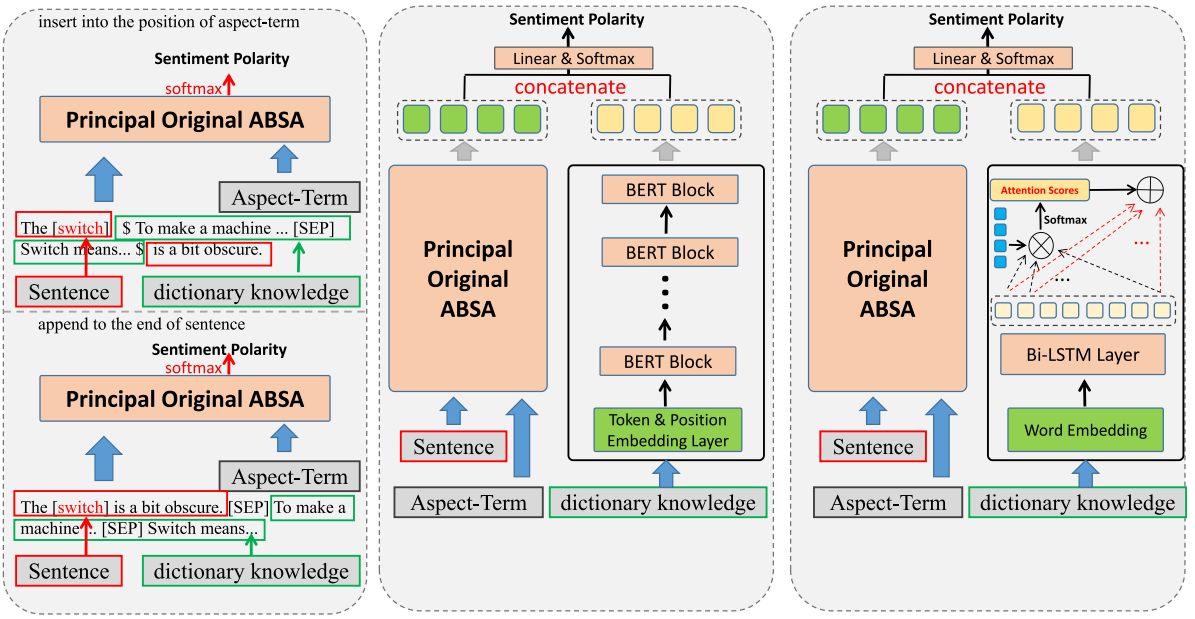
Sentence Concatenate (SENT-CONCAT) As shown at the top of Fig. 7(a), the concatenation strategy is the most intuitive and easy-to-understand approach, which is in compliance with the idea of NLP sequential modelling. However, based on this simple concatenation, the ABSA model cannot thoroughly understand the semantic relationship between the aspect term in the sentence and the added dictionary knowledge, which possibly results in incorrect reasoning. Inspired by Jian et al. (2020), to better leverage the retrieved knowledge, we believe that the position feature is also essential because the final ABSA performance depends on both its content information and its relative position information.

Sentence Insertion (SENT-INSERT) As shown at the bottom of Fig. 7(a), different from the concatenation strategy, the insertion strategy considers both the relative position and the absolute position. It incorporates the linearized knowledge into the sentence by inserting it after the aspect term. Furthermore, following Jian et al. (2020)'s experiment settings, we add the additional knowledge marker "$" to indicate the boundary of the descriptive knowledge (separate token "$" is inserted at the beginning and the ending position of the descriptive knowledge).

As shown in Fig. 7(b) and (c), there are two knowledge incorporation mechanisms on the feature encoder, namely BERT-based KI, and BiLSTM-Attn-based KI. Different from the token sequence-based knowledge incorporation discussed above, feature-encoder based knowledge incorporation requires an independent feature encoder to represent the implicit semantics of the retrieved descriptive knowledge. We introduce the two adopted feature encoders as follows.

Based on Transformers BERT Encoder Compared with the pre-trained embedding-based (e.g. Word2Vec, GloVe) RNNs (e.g. Bi-GRU, Bi-LSTM), which only provide context-independent representations, Transformers (e.g. BERT and its variants) is a revolutionary PLM technique which substantially improves the overall performance of the NLP tasks. Transformer BERT applies self-attention transformation to represent each input token in parallel, allowing for GPU's parallel calculation and generating a better contextualized representation. Thus, Transformers significantly improve the model efficiency and become more intelligent than RNNs for large-scale AI applications which have been quickly constructed within a short period.

Fig. 7(b) describes the details of the Transformer BERT-based knowledge infusion strategy, *BERT-based KI*. The Transformer-based language model BERT consists of N ($N \in \{8, 12, 16, \ldots\}$) stacked identical Transformer blocks. Each BERT block includes



(a) infusing knowledge to token sequence (two implementations) (b) infusing knowledge to feature encoder (BERT-based) (c) infusing knowledge to feature encoder (BiLSTM-Attention-based)

Fig. 7. Multiple alternative knowledge incorporation methods for the *Auxiliary Knowledge Injection* module in our experiments.

a standard multi-head self-attention network followed by a standard positional fully connected feedforward layer and a regular layer normalization. Here, we regard each BERT block as an independent module and denote the BERT block as $Transformer = \{TransBlock_L\}_1^N$, in which N represents BERT's depth. More specifically, given a retrieved dictionary knowledge sequence, we integrate the inputted vector sequence $\{h_{[CLS]}, h_1 \ldots, h_L\}$ as H_0, where the h_i is the ith token's initialized embedding vector provided by BERT in the sentence. Then the $L-layer$ transformer blocks transfer the word-level semantic feature representation by multiple layers. In an integral jth transformer blocks calculating procedure, the BERT hidden features H_j are computed using Eq. (8), as follows:

$$H^j = TransBlock_j(H^{j-1}) \tag{8}$$

where the $H^j = \{h^j_{[CLS]}, h^j_1 \ldots, h^j_L\}$ denotes the feature representations of the jth BERT block and the $H^{j-1} = \{h^{j-1}_{[CLS]}, h^{j-1}_1 \ldots, h^{j-1}_L\}$ denotes the feature representations of the $\{j-1\}$th BERT block. The BERT's output hidden state representations H^L can be seen as the contextualized representations of the input sentence. Then, we concatenate the knowledge's output representation and the *Principal Original ABSA* module's output representation. Finally, we would use the final concatenated representation to further downstream sentiment analysis.

Based on RNNs Encoder Note that simple RNN-based methods do not perform well in exploiting contextualized semantic features, especially for long-distance sentences. To avoid the vanishing-gradient problem (Hochreiter & Schmidhuber, 1997) in RNNs, we choose the Bidirectional LSTM (Bi-LSTM) as the primary encoder. Moreover, we argue that different words have different impacts on semantic recognition. Thus, following previous methods (Ding et al., 2022; Jin et al., 2022), to capture more valuable information, we apply the self-attention mechanism after the Bi-LSTM layer. Existing works (Tang, Qin, Feng et al., 2016) have demonstrated that the Bi-LSTM model with a self-attention mechanism achieves a better learning capability and leads to better performance than traditional RNN-based methods.

Fig. 7(c) describes the details of the *BiLSTM-Attn-based* knowledge infusion strategy, *BiLSTM-Attn-based KI*, which builds a standard self-attention network based on a standard Bi-LSTM layer in which we set the Bi-LSTM hidden size to 256. More specifically, given a dictionary descriptive knowledge sequence with L-length tokens, we firstly initialize *Word Embedding* layer by utilizing 300 dimensions GloVe embeddings[10] and use the pre-trained vectors to transfer the knowledge sequence to a vector sequence. After the word embedding step, the Bi-LSTM network provides a forward hidden state $\left(\overrightarrow{h_{[START]}}, \overrightarrow{h_1}, \overrightarrow{h_2}, \ldots, \overrightarrow{h_L}\right)$ and a backward hidden state $\left(\overleftarrow{h_{[START]}}, \overleftarrow{h_1}, \overleftarrow{h_2}, \ldots, \overleftarrow{h_L}\right)$. The Bi-LSTM feature extraction is computed via the following operations:

$$\mathbf{f}_j = \sigma\left(\mathbf{W}_{xf}\mathbf{x}_j + \mathbf{W}_{hf}\overleftarrow{\mathbf{h}_{j-1}} + \mathbf{b}_f\right) \tag{9}$$

[10] https://github.com/stanfordnlp/GloVe#download-pre-trained-word-vectors.

$$\mathbf{i}_j = \sigma\left(\mathbf{W}_{xi}\mathbf{x}_j + \mathbf{W}_{hi}\overrightarrow{\mathbf{h}_{j-1}} + \mathbf{b}_i\right) \tag{10}$$

$$\mathbf{o}_j = \sigma\left(\mathbf{W}_{xo}\mathbf{x}_j + \mathbf{W}_{ho}\overrightarrow{\mathbf{h}_{j-1}} + \mathbf{b}_o\right) \tag{11}$$

$$\mathbf{c}_j = \mathbf{f}_j \circ \mathbf{c}_{j-1} + \mathbf{i}_j \tanh\left(\mathbf{W}_{xc}\mathbf{x}_j + \mathbf{W}_{hc}\vec{\mathbf{h}}_{j-1} + \mathbf{b}_c\right) \tag{12}$$

$$\overrightarrow{\mathbf{h}_j} = \mathbf{o}_j \circ \tanh\left(\mathbf{c}_j\right) \tag{13}$$

In the above equations, the variable f_j is the input gate's activation vector; i_j is the forget gate's activation vectors; and o_j denotes the output gate's activation vector. The three gate representations f_j, i_j and o_j are generated through multiple intricate computations. The updated new memory of LSTM corresponds to the matrix multiplication of the input token feature x_j and the updating matrix W_{xf}, W_{xi} and W_{xo}. The remaining old memory of LSTM corresponds to the matrix multiplication of the last hidden state $\vec{\mathbf{h}}_{j-1}$ and the forgotten matrix W_{hf}, W_{hi} and W_{ho}. $\circ$ is the cell hidden state, σ is the sigmoid function, and $tanh$ is the hyperbolic tangent function. Finally, LSTM would convert the logical value into a probability value in $0 \sim 1$ based on an activation operation σ.

As we know, Bi-LSTM is the same as LSTM in the mechanical aspect. But when compared with vanilla LSTM, Bi-LSTM allows the information flow reversely in which the semantic feature can be calculated from the end position to the beginning position. After the semantic feature flows through Bi-LSTM, the forward hidden state $\overrightarrow{\mathbf{h}_j}$ and backward hidden state $\overleftarrow{\mathbf{h}_j}$ are concatenated. Then we can obtain the combined features $\mathbf{H} = \{h_{[START]}, h_1 \ldots h_j \ldots h_L\}$ where the jth hidden state $\mathbf{h}_j = [\overrightarrow{\mathbf{h}_j}; \overleftarrow{\mathbf{h}_j}]$. Next, the sequential hidden state representation $\mathbf{H}$ is fed into a standard self-attention layer as follows:

$$\alpha_j = \frac{\exp\left(a_j\right)}{\sum_{i=1}^{L}\exp\left(a_i\right)} \quad where \quad a_j = \tanh\left(\mathbf{W}^{\top}\left[\mathbf{h}_j\right] + b\right) \tag{14}$$

$$\mathbf{S} = \sum_j \alpha_j \mathbf{h}_j \quad where \quad \sum_j \alpha_j = 1 \quad \& \quad 0 < \alpha_j < 1 \tag{15}$$

As shown in Eq. (14), in the self-attention layer $\mathbf{W}$, a matrix multiplication layer, a *softmax* layer and an activation layer *tanh* will jointly calculate the attention weight α_j. Then, the final representation vector s_j which denotes the encoded dictionary knowledge, as shown in Eq. (15), is aggregated by the weighted sum operation of $\mathbf{h}_j$ and a_j. Finally, the obtained vector $\mathbf{S}$, which can be regarded as the final semantic of the external descriptive knowledge, would be concatenated with the output of the *Principal Original ABSA* module. Also, the concatenated result can be outputed to conduct the downstream ABSA task.

5. Experiment

Our paper focuses on investigating whether and how much external dictionary descriptive knowledge incorporation can improve ABSA's overall performance. In this section, our evaluation experiments are conducted based on the contrastive learning between our DictABSA and its original ABSA approaches, in which ten comparative ABSA approaches are adopted on five widely used benchmarks for the performance comparisons to demonstrate the superiority of our DictABSA.

5.1. Datasets and metrics

We firstly describe the five benchmarks used in the experiments and then briefly introduce the metrics we used to evaluate DictABSA.

5.1.1. Datasets

We conduct comprehensive experiments on five publicly available benchmark datasets, including Twitter,[11] SemEval 2014 (laptop & restaurant domain),[12] SemEval 2015 & SemEval 2016 (restaurant domain),[13] and ACLShortData.[14] These datasets are widely used in the ABSA task, which are adopted by most ABSA works. We carried out a large number of comparison experiments on these datasets. All the above datasets' aspect terms are labelled in three sentiment categories: neutral, positive, and negative.

Twitter: Twitter corpus is built by Mitchell et al. (2013), including approximately 10k English and 30k Spanish tweet comments, in which we utilize a part of the English language instances to conduct our experiments. Specifically, Li, Bing, Li et al. (2019) rearranged the Twitter dataset, and we use their dataset edition to train and test our models.

SemEval 2014: SemEval 2014 (Li, Bing, Zhang et al., 2019) is one officially released benchmark which is annotated from *SemEval-2014 Competition* (task 4), including the restaurant and laptop domains' reviews. Furthermore, we merge these two ABSA datasets into one complete SemEval2014 dataset for simplicity.

[11] Dataset is available at https://github.com/lixin4ever/E2E-TBSA/tree/master/data

[12] The task details can be accessed at http://alt.qcri.org/semeval2014/task4/.

[13] Dataset is available at https://github.com/lixin4ever/BERT-E2E-ABSA/tree/master/data.

[14] Dataset is available at https://github.com/songyouwei/ABSA-PyTorch/tree/master/datasets/acl-14-short-data.

Table 3
Statistics for the five benchmark datasets.

DataSet	Train				Test			
	POSITIVE	NEUTRAL	NEGATIVE	TOTAL	POSITIVE	NEUTRAL	NEGATIVE	TOTAL
Twitter	441	1129	117	1687	66	154	13	233
SemEval2014	1764	519	819	3102	644	187	180	1011
SemEval2015	574	27	149	750	244	25	131	400
SemEval2016	798	46	269	1113	321	25	74	420
ACLShortData	1403	2837	1383	5623	173	346	173	692

SemEval 2015 & SemEval 2016: SemEval 2015 and SemEval 2016 datasets (Li, Bing, Zhang et al., 2019) are annotated from the *SemEval-2015 Competition* (task 12) and the *SemEval-2016 Competition* (task 5). Compared with *SemEval 2014* benchmark, these two datasets are relatively small.

ACLShortData: ACLShortData is also a Twitter-based social dataset which is introduced by Dong et al. (2014) and annotated from *ACL' 14*. To effectively eliminate the impacts of the data imbalance problem, the corpora tweets are randomly sampled to make the data balanced. The neutral, negative, and positive categories respectively account for 50%, 25%, and 25%.

Followed by Li, Bing, Zhang et al. (2019), we apply the original dataset split strategy of 80%(training) and 20%(testing) in the same way since there is no standard train–test split scheme for these datasets. Apart from the different train–test split strategies, these original datasets are under no refactor in our experiments. As a result, it can better evaluate the ABSA models' real performance.

Table 3 reports the detailed statistics for the above five benchmarks, including the training instance count and test instance count of each benchmark's sentiment polarity, the training instance numbers, the test instance numbers, and the total instance numbers of each benchmark.

5.1.2. Metrics

We use the widely adopted evaluation metrics, f1-score and Accuracy to accurately evaluate the aspect-level sentiment analysis performance, which is same as previous methods to ensure comparability. Generally, a well-implemented ABSA approach always means a better performance gain, such as a higher $f1$ and a higher accuracy.

Accuracy is a standard assessment for measuring general tasks which can be simplified as label classification problems (e.g. ECI task Cao et al., 2021, ABSA task Ma et al., 2017). Accuracy denotes the percentage of correct predicted samples in all validation instances, which is defined as follows:

$$Accuracy = \frac{Count_T}{Count_N} \tag{16}$$

where the $Count_T$ is the correctly predicted sample number and $Count_N$ is the total sample number. For each evaluated instance, an ABSA prediction is predicted correctly only if the predicted sentiment polarity exactly matches its gold sentiment polarity. Otherwise, we regard it as wrong. As each ABSA sentence is annotated with a single sentiment polarity from three pre-defined aspect categories: POSITIVE/NEUTRAL/NEGATIVE, it can also be especially useful to study the fine-grained sentiment analysis task in our work.

The f1-score is also a typical classification metric which is adopted for various tasks and datasets. It is equivalent to the harmonic mean of the precision and recall, where the best value is 1.0, and the worst value is 0.0. In this case, the metric f1-scores can be used to determine which one produces better classification results. The formula for the f1-score is:

$$f1 = \frac{2 * Prec * Rec}{Prec + Rec} \qquad where \quad Prec = \frac{TP}{TP + FP} \quad \& \quad Rec = \frac{TP}{TP + FN} \tag{17}$$

where the $Prec$ and Rec denote the precision rate and the recall rate of the label classification model; TP (True Positive) is the number of samples when the actual class is yes and the predicted class is also yes; FP (False Positives) is the number of samples when the actual class is no and the predicted class is yes; FN (False Negative) is the number of samples when the actual class is yes but the predicted class in no.

5.2. Experiment setting

Hyperparameter settings have a significant impact on the experiment outcome. Therefore, all the optimal hyperparameter settings in this paper have been validated by a large number of comparative experiments. We summarize these as follows.

We use the pre-trained word vectors released by *StanfordNLP GloVe*[15] to initialize the adopted RNN-based models' word embeddings. Specifically, for all RNN-based models evaluated on the Twitter dataset, we apply the pre-trained vectors "GloVe-Twitter"[16] (2B tweets, 27B tokens, 1.2M vocab, uncased, 200 dimensions) which is trained on a Twitter corpus. For all RNN-based models evaluated on the non-Twitter dataset, we apply the "GloVe-CommonCrawl"[17] (42B tokens, 1.9M vocab, uncased, 300

[15] https://github.com/stanfordnlp/GloVe#download-pre-trained-word-vectors.
[16] https://huggingface.co/stanfordnlp/glove/resolve/main/glove.twitter.27B.zip.
[17] https://huggingface.co/stanfordnlp/glove/blob/main/glove.42B.300d.zip.

Table 4
Optimal training time hyper-parameter settings for all adopted DictABSA variants on each knowledge infusion strategy.

Model		Epoch	Batch size	Learning rate	Patience	Max Sent length	Embed dim	Hidden dim
SENT-CONCAT	RNN-based	100	16	2e−05	50/30	125	200/300	300
	BERT-based	50	16	2e−05	20/5	125	768	768
	ABSA-DeBERTa	32	16	1e−05	–	256	768	768
SENT-INSERT	RNN-based	100	16	2e−05	50/30	125	200/300	300
	BERT-based	50	16	2e−05	20/5	125	768	768
	ABSA-DeBERTa	32	16	1e−05	–	256	768	768
RNN-based encoder	RNN-based	100	16	2e−05	50/30	125	200/300	300
	BERT-based	50	16	2e−05	20/5	125	768	768
	ABSA-DeBERTa	32	16	1e−05	–	256	768	768
Transformer BERT-based encoder	RNN-based	100	16	2e−05	50/30	125	200/300	300
	BERT-based	50	16	2e−05	20/5	125	768	768
	ABSA-DeBERTa	32	16	1e−05	–	256	768	768

dimensions) which is trained on massive web corpora (e.g. Wikipedia, Gigaword). The embeddings of the out-of-vocabulary words will be all zeros.

For all pre-trained BERT-based models, the BERT encoder is implemented as standard HuggingFace BERT architecture: Bert-Base-Uncased[18], which does not detect any difference between uppercase letters and lowercase letters. Bert-Base-Uncased contains 12 transformer blocks, 768 hidden dimensions, 12 multi-head self-attentions, and about 110M parameters. To avoid the GPU RAM's out-of-memory issue, we give up adopting BERT-Large (more than 340M parameters) even through BERT-Large could result in a better performance. More model versions of BERT (e.g. BERT-Mini uncased, BERT-Large cased) can be downloaded from this repository.[19] The pre-trained BERT, Bert-Base-Uncased, which contains 12 transformer blocks and hidden dimension which is set to 768, is adopted as the BERT encoder.

To evaluate the precise and accurate difference in the DictABSA performance, after careful consideration, we take strict external hyperparameter control over each DictABSA model training process on each dataset. For all DictABSA variants, the Adam optimizer (Kingma & Ba, 2014) is utilized intuitively, the dropout rate dr is set to 0.1 for all models, and the Xavier initialization is adopted for all trainable network parameters (e.g. Bi-LSTM, full connect, and attention layer) in addition to BERT transformer. As we know, the regularization instability between network layers would reduce the final performance of the models. To address this challenge, the batch size is set to 16 for each iteration. To better reserve the context semantics, we limit the maximum length of sentences to 125 words, which includes the original sentence and the dictionary descriptive knowledge of its aspect term. In the objective function, we adopt weight decay as the regularization method rather than L_2 normalization, and the weight decay rate is set to 0.01 to avoid overfitting. Additionally, we also adopt the early-stopping strategy by evaluating Accuracy per 20 training epochs on the test set to avoid overfitting.

We choose the best result to represent the final model performance for comparison due to the random fluctuation of the experimental results. Lastly, considering the reproducibility of our experiments, we set the random seed to a fixed number: 1234 for generating deterministic numbers rather than random numbers.

Furthermore, we list the detailed statistics of the optimal training time hyper-parameters we adopted in order to provide readers with concise yet comprehensive insights into our experiments.

Table 4 summarizes the statistical information as follows. The first column details the four adopted knowledge infusion strategies and the second column details the three Principal ABSA methods, namely RNN-based methods (i.e. LSTM, ATAE-LSTM, TD-LSTM, TC-LSTM, IAN, MemNet and CABASC), BERT-based methods (i.e. BERT-SPC, LCF-BERT-ABSA) and ABSA-DeBERTa. The setting of the training patience for RNN-based methods includes two options, 50 epochs and 30 epochs and the patience setting for BERT-based methods also includes two options, 20 epochs and 5 epochs. For DictABSA which adopts ABSA-DeBERTa, there is no training patience. The dimension of word embedding trained is set to 200 and 300 for the reason that we simultaneously adopt the pre-trained vectors GloVe-Twitter on the Twitter dataset and adopt the pre-trained vectors GloVe-CommonCrawl on the non-Twitter datasets.

5.3. Principal ABSA methods

The experimental ABSA methods which are adopted as the *Principal Original ABSA* module are briefly described as follows:

- **LSTM:** LSTM (Hochreiter & Schmidhuber, 1997) is a special kind of RNN which is composed of a cell, an input gate, an output gate, and a forget gate. Compared with standard RNNs, LSTM is explicitly designed to alleviate the long-term dependencies (LTD) problem that existed in vanilla RNNs.
- **ATAE-LSTM:** ATAE-LSTM (Wang et al., 2016) is an attention mechanism based on LSTM, which explicitly explores aspect term-specific context semantics by appending the aspect term embedding to each word embedding of the input sentence.

[18] https://huggingface.co/bert-base-uncased.
[19] https://huggingface.co/transformers/installation.html#offline-mode.

Table 5
The experiment performance statistics (% Accuracy) of the ABSA baselines without incorporating descriptive knowledge.

Models & DataSet	Twitter	SemEval 2014	SemEval 2015	SemEval 2016	ACLShortData
LSTM	0.7417	0.6798	0.8293	0.8220	0.6560
TD-LSTM	0.7226	0.7203	0.8354	0.8155	0.6846
TC-LSTM	0.7194	0.7042	0.8481	0.8252	0.7019
ATAE-LSTM	0.7164	0.7052	0.8415	0.8305	0.7104
IAN	0.7232	0.7021	0.8293	0.8390	0.6944
MEMNET	0.7518	0.6991	0.8659	0.8305	0.6928
CABASC	0.7032	0.7074	0.8354	0.7670	0.6327
BERT-SPC	0.7607	0.7825	**0.9046**	0.8729	0.7088
LCF-BERT-ABSA	**0.7661**	0.8085	0.9024	0.8898	0.7216
ABSA-DeBERTa	0.7512	**0.8187**	0.8846	**0.9158**	**0.7408**

- **TD-LSTM:** TD-LSTM (Tang, Qin, Feng et al., 2016) is an LSTM-based architecture that can obtain context features from both left and right sides by dividing the input sequences into left context and right context towards a specific aspect term.
- **TC-LSTM:** TC-LSTM (Tang, Qin, Feng et al., 2016) is a relatively simple LSTM-based network which employs concatenation and other strategies to fuse the aspect information with the sentence context.
- **IAN:** IAN (Ma et al., 2017) is an interactive attention network which makes use of two LSTM-based attention layers to model target and context independently and generates the interactive representations for both target and context.
- **MemNet:** MemNet (Tang, Qin, Liu, 2016) is a deep memory network which consists of multiple computational layers. Each computational layer is a neural attention model over an external memory.
- **CABASC:** CABASC (Liu et al., 2018) is a novel content attention-based ABSA model with two attention enhancing mechanisms: sentence-level content attention mechanism and the context attention mechanism, to deal with the syntactical structure of the complex sentences.
- **BERT-SPC:** BERT-SPC (Devlin et al., 2019) is a sentence pair classification task of the pre-trained BERT model which uses the delimiter [SEP] to combine the sentence with the aspect term.
- **LCF-BERT-ABSA:** LCF-BERT-ABSA (Zeng et al., 2019) is a local context focus (LCF) mechanism based on a BERT-shared layer. It utilizes the Context features Dynamic Mask and Context Features Dynamic Weighted layers to capture the long-term internal dependencies of sentence contexts and aspect terms.
- **ABSA-DeBERTa:** ABSA-DeBERTa (Emanuel & Marcacini, 2021) is a decoder-enhanced BERT with disentangled attention which brings a considerable improvement to the ABSA task by introducing two novel techniques: a disentangled attention mechanism and an enhanced mask decoder.

5.4. Experimental results

We conduct a series of in-depth experimental analyses on the five benchmark datasets to validate the effectiveness and superiority of our DictABSA, and the following questions are answered:

Question 1: Can the dictionary descriptive knowledge enhance the performance of the ABSA task? How effective is the proposed DictABSA in improving language understanding ability (sentiment analysis) when compared with other traditional approaches?

Question 2: Do the different dictionary knowledge infusion strategies directly influence the final sentiment analysis performance? From a quantitative point of view, to what degree do our proposed knowledge infusion strategies increase the final model performance?

5.4.1. Original ABSA performance analysis (baselines)

To illustrate experimental comparability between DictABSA and non-DictABSA better, we first conduct a series of experiments in relation to original ABSA approaches across the five benchmark datasets as the baseline performances, which do not incorporate dictionary descriptive knowledge.

As shown in Table 5, it is clear that in terms of Accuracy, the performances of the traditional RNN-based ABSA methods are overall worse than the Accuracy obtained by the pre-trained Transformer BERT-based counterparts, which is as we expected. The performance comparisons between the BERT and non-BERT methods verify the widely recognized fact that the multi-head self-attention mechanism inside BERT is extremely helpful for the natural language understanding task. Compared with all the RNN-based methods, our adopted BERT-based ABSA methods achieve the best and second-best Accuracy score while the RNN-based methods obtain inferior performance. For example, as presented in Rows. 9 and 10 in Col. 5, which are shown in the second group, *BERT-SPC* obtains the best Accuracy of 90.46%, and *LCF-BERT-ABSA* obtains the second-best Accuracy of 90.24%.

Furthermore, from the first group in Table 5, we can see that the performance of non-BERT-based models is not very stable. For example, as for the *TD-LSTM* and *MEMNET* models evaluated on the Twitter and SemEval 2014 benchmarks, *MEMNET* obtains an Accuracy of 75.18% which is a 2.92% gain compared with *TD-LSTM* on Twitter. But when evaluating the SemEval 2014 benchmark, we note that the *TD-LSTM* has an inferior performance of 69.91% Accuracy, which is an obvious performance drop of 1.2% in Accuracy compared with the Accuracy of *TD-LSTM*, 73.03%. We attribute this to the poor domain transferability of the fixed

Table 6
The statistics of the DictABSA performance, which adopts the sentence concatenate strategy, corresponding to Section 5.4.2. The black-coloured values denote the absolute metric of Accuracy, which are reported using (%). The green-coloured and red-coloured numbers denote the increased and decreased Accuracy compared with the baseline performance.

Models & DataSet	Twitter	SemEval 2014	SemEval 2015	SemEval 2016	ACLShortData
LSTM	0.6848 ↓ −5.69%	0.6770 ↓ −0.28%	0.8293 ↓ 0.00%	0.7288 ↓ −9.32%	0.6256 ↓ −3.04%
TD-LSTM	0.6848 ↓ −3.78%	0.7138 ↓ −0.65%	0.8354 ↓ 0.00%	0.8058 ↓ −0.97%	0.6976 ↑ 1.30%
TC-LSTM	0.7078 ↓ −1.16%	0.6913 ↓ −1.29%	0.8354 ↓ −1.27%	0.8252 ↓ 0.00%	0.7180 ↑ 1.61%
ATAE-LSTM	0.7203 ↑ 0.39%	0.7082 ↑ 0.30%	0.8537 ↑ 1.22%	0.8475 ↑ 1.7%	0.7280 ↑ 1.76%
IAN	0.7351 ↑ 1.19%	0.6383 ↓ −6.38%	0.8293 ↓ 0.00%	0.7288 ↓ −11.02%	0.5408 ↓ −15.36%
MEMNET	0.7542 ↑ 0.24%	0.7082 ↑ 0.91%	0.8537 ↓ −1.22%	0.8305 ↓ 0.00%	0.7136 ↑ 2.08%
CABASC	0.7259 ↑ 2.27%	0.6977 ↓ −0.97%	0.8354 ↓ 0.00%	0.7947 ↑ 1.77%	0.6419 ↑ 0.92%
BERT-SPC	0.7796 ↑ 1.89%	0.8250 ↑ 4.25%	**0.9390** ↑ 3.44%	0.9237 ↑ 5.08%	0.7504 ↑ 4.16%
LCF-BERT-ABSA	**0.7852** ↑ 1.91%	0.8237 ↑ 1.52%	0.9263 ↑ 2.39%	0.9153 ↑ 2.55%	0.7488 ↑ 2.72%
ABSA-DeBERTa	0.7583 ↑ 0.72%	**0.8610** ↑ 4.23%	0.9024 ↑ 1.78%	**0.9492** ↑ 3.34%	**0.7872** ↑ 4.64%

pre-trained embeddings and the training instability of RNN-based models. More specifically, a small hyper-parameter modification (e.g. dropout rate, batch size, and learning rate) may bring about a large performance fluctuation. The fixed pre-trained embeddings, which contain the restaurant domain knowledge, are not suitable for the social interaction field. In contrast, using the learned commonsense knowledge inside BERT during the large-scale corpora pre-training procedure, BERT-based ABSA approaches always retain relatively good robustness during training optimization, regardless of the variations in various hyper-parameters.

In the following sections, the comparative experiments which corresponds to the proposed knowledge-infusion strategies will take the baseline experimental performance shown in Table 5 as comparison baselines, to better demonstrate the superiority of incorporating external dictionary descriptive knowledge.

5.4.2. DictABSA analysis: Sentence concatenate strategy

As illustrated in Table 6, we evaluate the performance of DictABSA, which adopts the simple natural language sequence concatenate strategy between the sentence and its aspect term's dictionary descriptive knowledge.

As shown in the last two row-groups of Table 6, from the performance of the dictionary knowledge-enhanced BERT-based models and the BERT-based DictABSAs, we can see that the dictionary noun-explanations substantially improve the Accuracy absolute gains on several of the benchmarks by about 1%–3%, suggesting that infusing external commonsense knowledge such as dictionary explanations is extremely helpful for most original Transformer-based models, leading to significant improvement on the ABSA task.

From the first row-group of Table 6, the experiment results for RNN-based DictABSAs are worse than the baselines' performance compared with the Transformer-based methods, showing that the strategy of concatenating sentences with dictionary knowledge may not work well in this situation. Intuitively, we know that RNNs are not capable of handling the long-term dependencies (LTD) problem (Hochreiter & Schmidhuber, 1997). LTD is an typical issue that exists in gradient-descent learning algorithms for RNNs, causing that when learning (forwarding) the remainder context of an extremely long sentence, it has been quite difficult to completely remember (also easy to forget) the previous semantic features.

The weak performance of the RNN-based models may be because the final model inputs become much longer than the original sentences after concatenating the noun-description knowledge, especially when the infused noun-descriptions are extremely long. Especially for IAN, the performance of IAN drops significantly decrease by up to 11% and 15% in absolute Accuracy. The designed interactive attention mechanism is not suitable for the simple sentence concatenate strategy of the external knowledge infusion mechanism. More specifically, the *Average/Max Pooling* operations inside the interactive attention mechanism of IAN probably lose more original context information in modelling the long sentence after knowledge concatenation. This may be the primary reason why the IAN network performs worse than most RNN-based models.

As for the BERT-based DictABSAs, including BERT-SPC, LCF-BERT-ABSA, and ABSA-DeBERTa, these models perform better than all the datasets compared with the BERT-based baselines. Significantly, the BERT-SPC and ABSA-DeBERTa models achieve the best Accuracy performance of 86.10% and second-best Accuracy score of 82.50%, which outperforms the baselines by 4.23% and 4.25% in terms of absolute Accuracy gains, respectively. After further investigation, it is found that the inter-architecture of the Transformer can better capture the long-term dependencies than the traditional RNN-based models to result in a significant performance improvement, which indicates that the multi-head self-attention plays an important role in long-term semantic feature extraction. In other words, Transformer (e.g. BERT), a novel high-parallel semantic encoding architecture, can effectively resist the phenomenon of long-term semantic dissipation existing in RNNs and have a better semantic understanding capability when handling sentences of long lengths.

To further investigate the reason why RNN-based ABSAs achieve inferior performance, we counted the average length of the original sentence and its corresponding added knowledge to gather statistics for each benchmark, which is shown in Fig. 8. From each benchmark's length statistics shown in the table, the sentence of retrieved knowledge is two to three times the length of the original sentence on average. For example, in the training set of the SemEval 2014 benchmark, the average sentence length of its dictionary knowledge is 38.39 compared with the before-concatenation sentence length of 15.97, which is about 2.4 times the length of the original one. The severe disproportion between the volumes of expressions and dictionary knowledge can be explicitly observed, which might be the key reason for the inferior performance of most RNN-based ABSAs.

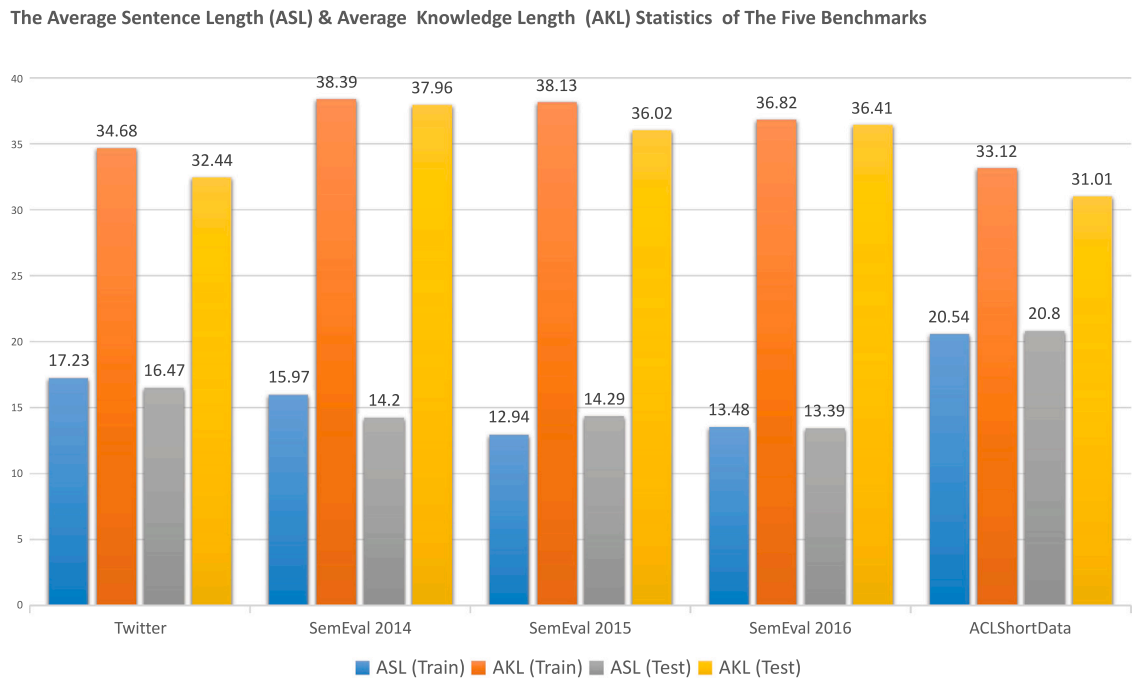


Fig. 8. The statistics for the length of the original sentence and its descriptive knowledge of the five used benchmarks (Unit: Word, Punctuation, and other special tokens).

As shown in the last column of Table 6, we can see that when compared with the baselines on the ACLShortData benchmark, the performance of the RNN-based ABSAs on the ACLShortData, such as TC-LSTM and MEMNET, achieves a competitive improvement in Accuracy in the range of 0.92% to 2.08%. At the same time, we can observe that the average length ratio (ASL:AKL) between the original expressions and dictionary knowledge which is about (≈1:1.5) is relatively reasonable. The experiments reveal that, for the situation of infusing length-balanced knowledge, the RNN-based ABSAs can still perform well to a certain degree. In the next several sections, we investigate various of solutions for improving the descriptive knowledge infusion strategy.

Although most RNN-based DictABSAs achieve worse performance than the baselines without knowledge infusion, the long-term dependencies problem does not seem to affect the final performance of these models (ATAE-LSTM, MEMNET, and CABASC), as shown in some fields of the first group in Table 6. For example, ATAE-LSTM still outperforms the baselines, which is a 1.2% absolute Accuracy gain across all the five benchmarks on average. Probably the proposed self-attention mechanism inside the ATAE-LSTM makes the model not easily forget the learned context characteristics while learning the extremely long dictionary descriptive knowledge. This helps to extract the sentence's long-term semantic features for the final vector representation to better conduct semantic-based sentiment identification. Also, compared with the baselines, MEMNET is slightly improved on the Twitter, SemEval 2014, and ACLShortData datasets, which is a 1.2% Accuracy gain over the baselines on average. We attribute this to the superior long-term information memory power of the multiple deep memory network layers inside the MEMNET. As a result, the multiple memory units and the attention mechanism both mean MEMNET's final performance is not susceptible to the effect of significantly increased sentence length.

5.4.3. DictABSA analysis: Knowledge Description Insertion Strategy

We found that the direct sentence concatenation strategy easily makes the DictABSAs ignore the important relative position information between aspect terms and their dictionary explanations. In addition to the simple sentence concatenation-based strategy, we also explore the sentence insertion strategy (inserting the noun-descriptions into the sentence after the aspect term), which considers the importance of the relative position. More details are provided in the *Knowledge Injection (2)* part of Section 4.4. The following observations can be made about the performance of the Knowledge Description Insertion Strategy-enhanced DictABSAs as detailed in Table 7: (1) ***Effectiveness of Knowledge Insertion Strategy:*** As illustrated in the last two row-groups of Table 7. The Sentence Insertion Strategy-enhanced DictABSAs, which is based on the pre-trained BERT (e.g. BERT-SPC, LCT-BERT-ABSA, and ABSA-DeBERTa), is much greater than that of the sentence concatenation-based. For example, the performance with LCF-BERT-ABSA among most benchmarks is much better than that of the baselines without knowledge infusion (by 3.1% absolute Accuracy gains on average). We can see that the performance of the insertion strategy based on LCF-BERT-ABSA is better than that of the concatenate strategy based. As shown in the second-last row of Table 7, the Accuracy of the Knowledge Insertion Strategy-based DictABSA respectively achieves 1.82%, 2.49%, 1.25%, and 1.56% absolute gains on the SemEval2014/2015/2016 and ACLShortDarta datasets than the corresponding results which are shown in Table 6. These overall experiment results prove that the adopted Knowledge

Table 7
The statistics of the DictABSA performance, which adopts the Knowledge Description Insertion Strategy, corresponding to Table 7. The green-coloured and red-coloured numbers denote the increased and decreased Accuracy compared with the baseline performance, respectively.

Models & DataSet	Twitter	SemEval 2014	SemEval 2015	SemEval 2016	ACLShortData
LSTM	0.7232 ↓ −1.85%	0.7112 ↑ 3.14%	0.8293 ↓ 0.00%	0.7288 ↓ −9.32%	0.6976 ↑ 4.16%
TD-LSTM	0.7242 ↑ 0.16%	0.7106 ↓ −0.97%	0.8354 ↓ 0.00%	0.8155 ↓ 0.00%	0.6531 ↓ −3.15%
TC-LSTM	0.7078 ↓ −1.16%	0.6817 ↓ −2.25%	0.8354 ↓ −1.27%	0.7767 ↓ −4.85%	0.6531 ↓ −4.88%
ATAE-LSTM	0.7255 ↑ 0.91%	0.6960 ↓ −0.92%	0.8415 ↓ 0.00%	0.8356 ↓ 0.51%	0.7248 ↑ 1.44%
IAN	0.7136 ↓ −0.96%	0.6535 ↓ −4.86%	0.7835 ↓ 4.58%	0.7288 ↓ −11.02%	0.5408 ↓ −15.36%
MEMNET	0.7422 ↓ −0.96%	0.6991 ↓ 0.00%	0.8537 ↓ −1.22%	0.8212 ↓ −0.93%	0.7120 ↑ 1.92%
CABASC	0.7199 ↑ 1.67%	0.6752 ↓ −3.22%	0.8354 ↓ 0.00%	0.7379 ↓ −2.91%	0.6252 ↓ −0.75%
BERT-SPC	**0.7885** ↑ 2.78%	0.8393 ↑ 5.68%	0.9294 ↑ 2.48%	0.9153 ↑ 4.24%	0.7568 ↑ 4.8%
LCF-BERT-ABSA	0.7852 ↑ 1.91%	0.8419 ↑ 3.34%	**0.9512** ↑ 4.88%	0.9278 ↑ 3.80%	0.7644 ↑ 4.28%
ABSA-DeBERTa	0.7701 ↑ 1.89%	**0.8610** ↑ 4.23%	0.9146 ↑ 3.00%	**0.9322** ↑ 1.64%	**0.7801** ↑ 3.93%

Insertion Strategy-enhanced DictABSA has certain advantages compared with the baselines and the Concatenation Strategy-based DictABSA.

(2) ***Still Existed Long-Term Dependencies:*** Due to the Knowledge Insertion Strategy which greatly increases the language sequence length, the long-term dependencies problem is also not effectively solved in the RNN-based DictABSAs. As shown in the first group in Table 7, similar observations are captured in the comparison between the non-BERT DictABSAs' performances with the baselines (Section 5.4.1). Moreover, IAN still performs worse than most of the other DictABSA because of its *Average/Max Pooling*-based interactive attention mechanism. Traditional knowledge augmentation methods of infusing during sequence generation cannot fundamentally solve the long-term dependencies issue. Thus, it is necessary to explore more powerful infusion strategies for the auxiliary *Knowledge Injecting* module to better represent the added dictionary knowledge. The next sections introduce corresponding proposed strategies for solving the long-term dependencies problem.

Furthermore, from the last three rows in the second column of Tables 5 and 7, we can observe that the statistics of the BERT-based DictABSAs' performance on the Twitter dataset are not as good as its performance on the other four datasets. To be specific, compared with the baselines (Section 5.4.1), BERT-based DictABSA, such as LCF-BERT-ABSA and ABSA-DeBERTa only increase 1.91% and 1.89% in absolute Accuracy on the Twitter benchmark while increasing 4.28% and 3.93% in absolute Accuracy on the ACLShortData benchmark. After further analyses, we conclude that this may be because the Twitter benchmark consists of a set of Internet comments (tweets), and there exists many misspelled phrases and informal linguistic grammar which is not recorded in the Oxford Dictionary. Most tweets contain many grammatically incorrect comments as well as informal daily-life expressions and abbreviations commonly used on the Internet, causing difficulty when extracting high-quality and accurate noun-descriptions from the Oxford Dictionary API. The retrieved inaccurate commonsense knowledge has much semantic noise, so that it cannot bring improvements as significant as that of other benchmarks to the ABSA.

5.4.4. DictABSA analysis: Bi-LSTM with Attention-based Knowledge Infusion Strategy

Table 8 compares the performance of our RNN and attention-based Knowledge Infusion Strategy based DictABSAs against the baselines without knowledge infusion. Our DictABSAs, which utilizes an independent Bi-LSTM with an attention network to embed the dictionary knowledge, overwhelmingly outperform all baselines (Section 5.4.1) in terms of the Accuracy evaluation metric on the five benchmarks. For the results of the BERT-based DictABSAs shown in the last two row-groups of Table 8, after being equipped with Bi-LSTM with the attention network as the knowledge encoder, these models obtain Accuracy absolute gains of about 2% to 6% compared with the baselines.

For RNN-based DictABSAs (non-BERT), as presented in the first row-group of Table 8, these DictABSAs do not obtain inferior performance similar to the previous concatenation or insertion strategies but obtain overall performance improvements across the Accuracy metric, such as a 3.79% absolute Accuracy gain for LSTM on Twitter and a 2.88% absolute Accuracy gain for LSTM on ACLShortData. These experiment results powerfully demonstrate that the superior extra-deep neural network (extra-DNN) encoding method for external dictionary knowledge infusion can effectively solve the long-term dependencies problem and gain a better global context semantic representation after concatenating dictionary knowledge embedding. Thus, the overall ABSA performance is significantly improved due to the better semantic vector representations which are used to classify the final sentiment polarity.

For BERT-based DictABSAs, as shown in the last two row-groups of Table 8, all BERT-based DictABSAs simultaneously outperform the baselines and the previous two strategies, achieving about a 3% absolute gain over the baselines and more than a 1% absolute gain over the above strategies on average across the Accuracy metric. Overall, the Bi-LSTM with the attention-based Knowledge Infusion Strategy is much more helpful than the baselines, including the above two simple strategies, reflecting the importance of the different adopted strategies of the knowledge infusion mechanism. In Section 5.4.5, we explore a more robust knowledge infusion strategy which leverages the Transformer BERT as the dictionary knowledge semantic encoder to further enhance our DictABSA's final performance.

Table 8
The performance statistics of the DictABSAs, which adopts the Bi-LSTM with Attention-based Knowledge Infusion Strategy, corresponding to Section 5.4.4. The green-coloured numbers denote the increased Accuracy compared with the baselines.

Models & DataSet	Twitter	SemEval 2014	SemEval 2015	SemEval 2016	ACLShortData
LSTM	0.7796 ↑ 3.79%	0.7114 ↑ 3.16%	0.8391 ↑ 0.98%	0.8456 ↑ 2.36%	0.7074 ↑ 5.14%
TD-LSTM	0.7406 ↑ 1.80%	0.7395 ↑ 1.92%	0.8422 ↑ 0.68%	0.8319 ↑ 1.64%	0.7107 ↑ 2.61%
TC-LSTM	0.7341 ↑ 1.47%	0.7276 ↑ 2.34%	0.8613 ↑ 1.32%	0.8537 ↑ 2.85%	0.7251 ↑ 2.32%
ATAE-LSTM	0.7351 ↑ 1.87%	0.7183 ↑ 1.31%	0.8563 ↑ 1.48%	0.8462 ↑ 1.57%	0.7446 ↑ 3.42%
IAN	0.7507 ↑ 2.75%	0.7080 ↑ 0.59%	0.8323 ↑ 0.30%	0.8579 ↑ 1.89%	0.7129 ↑ 1.85%
MEMNET	0.7882 ↑ 3.64%	0.7157 ↑ 1.66%	0.8957 ↑ 2.98%	0.8411 ↑ 1.06%	0.7298 ↑ 3.70%
CABASC	0.7327 ↑ 2.95%	0.7191 ↑ 1.17%	0.8504 ↑ 1.50%	0.7943 ↑ 2.73%	0.6615 ↑ 2.88%
BERT-SPC	0.7885 ↑ 2.78%	0.8443 ↑ 6.18%	0.9432 ↑ 3.86%	0.9225 ↑ 4.96%	0.7624 ↑ 5.36%
LCF-BERT-ABSA	**0.7924** ↑ 2.63%	0.8449 ↑ 3.64%	**0.9438** ↑ 4.14%	0.9304 ↑ 4.06%	0.7704 ↑ 4.88%
ABSA-DeBERTa	0.7779 ↑ 2.67%	**0.8696** ↑ 5.09%	0.9068 ↑ 2.22%	**0.9498** ↑ 3.40%	**0.7864** ↑ 4.56%

Table 9
The performance statistics of the DictABSAs which adopts the BERT-based Knowledge Infusion Strategy, corresponding to Section 5.4.4. The green-coloured numbers denote the increased Accuracy compared with the baselines.

Models & DataSet	Twitter	SemEval 2014	SemEval 2015	SemEval 2016	ACLShortData
LSTM	0.7725 ↑ 3.08%	0.7244 ↑ 4.46%	0.8461 ↑ 1.68%	0.8424 ↑ 2.04%	0.7182 ↑ 6.22%
TD-LSTM	0.7394 ↑ 1.68%	0.7529 ↑ 3.26%	0.8464 ↑ 1.10%	0.8357 ↑ 2.02%	0.7111 ↑ 2.65%
TC-LSTM	0.7520 ↑ 3.26%	0.7314 ↑ 2.72%	0.8866 ↑ 3.85%	0.8741 ↑ 4.89%	0.7289 ↑ 2.70%
ATAE-LSTM	0.7483 ↑ 3.19%	0.7235 ↑ 1.83%	0.8667 ↑ 2.52%	0.8752 ↑ 4.47%	0.7398 ↑ 2.94%
IAN	0.7561 ↑ 3.29%	0.7272 ↑ 2.51%	0.8465 ↑ 1.72%	0.8541 ↑ 1.51%	0.7360 ↑ 4.16%
MEMNET	0.7716 ↑ 1.98%	0.7111 ↑ 1.20%	0.8833 ↑ 1.74%	0.8457 ↑ 1.52%	0.7202 ↑ 2.74%
CABASC	0.7299 ↑ 2.67%	0.7346 ↑ 2.72%	0.8536 ↑ 1.82%	0.7812 ↑ 1.42%	0.6597 ↑ 2.70%
BERT-SPC	0.7987 ↑ 3.80%	0.8467 ↑ 6.42%	0.9362 ↑ 3.16%	0.9253 ↑ 5.24%	0.7678 ↑ 5.90%
LCF-BERT-ABSA	**0.7930** ↑ 2.69%	0.8455 ↑ 3.70%	**0.9484** ↑ 4.60%	0.9360 ↑ 4.62%	0.7792 ↑ 5.76%
ABSA-DeBERTa	0.7847 ↑ 3.36%	**0.8760** ↑ 5.73%	0.9273 ↑ 4.27%	**0.9530** ↑ 3.72%	**0.7934** ↑ 5.26%

5.4.5. DictABSA analysis: BERT-based Knowledge Infusion Strategy

Note that the natural language modelling capability of the pre-trained Transformers is far superior to traditional fixed embeddings (e.g. Word2Vec, GloVe). To further enhance our knowledge infusion mechanism, we substitute the RNN-attention layer with the Transformer-based language representation model; specifically, BERT (Yang et al., 2022), for its superior performance to existing methods. The specific model implementations are called BERT-based Knowledge Infusion Strategy-based DictABSAs. Table 9 shows the experimental results of the DictABSAs based on the BERT-based Knowledge Infusion Strategy.

From Table 9, we can observe that regardless of the RNN-based DictABSA (non-BERT) or BERT-based DictABSA, compared to the above strategies, there are overall considerable improvements in the final model performance across the Accuracy metric. For example, ATAE-LSTM (non-BERT) and BERT-SPC (BERT) under the BERT-based Knowledge Infusion Strategy are superior to that of Bi-LSTM with Attention-based Knowledge Infusion Strategy, as we see an average 1.06% and 0.28% performance gains, respectively. These discoveries confirm our hypothesis that adopting a per-trained Transformer as our dictionary knowledge encoder greatly contributes to the overall performance of ABSA. Limited by the computing capability of the GPU, we only consider the standard pre-trained BERT Bert-Base-Uncased to encode the retrieved dictionary knowledge in our experiments whereas trying more superior Transformers (e.g. BERT-Large, RoBERTa) may additionally contribute to the final ABSA performance.

In conclusion, despite different implementations, the knowledge injection module always achieves a more competitive performance than the baselines, which conclusively demonstrates the general applicability and effectiveness of our DictABSA. In addition to the four knowledge infusion strategies proposed in this paper, there are still many valuable research directions to exploit more powerful knowledge-infusion approaches to better leverage external knowledge, or other types of commonsense knowledge.

6. Case analysis

To better understand our DictABSA design, we also conduct case analyses to further verify the contributions and effectiveness of the dictionary knowledge-infusion strategies systematically. Table 10 shows several cases of the IAN-based DictABSA and the DeBERTa-based DictABSA under all the proposed knowledge infusion strategies.

From the sentiment analysis case visualizations in Table 10, several important facts can be observed:

(i) The overall results demonstrate the effectiveness of our research innovations on dictionary knowledge enhancement. The baselines fail to correctly identify the semantics from the single sentences and make false sentiment predictions. In contrast, our proposed dictionary knowledge infused ABSA, DictABSA, is able to recognize the correct sentiment polarities, especially those DictABSAs equipped with independent knowledge encoders. This illustrates the superiority of our DictABSA due to its interactive features between the contextualized semantics and the infused background knowledge.

(ii) As observed in the sixth row of Table 10, when equipped with sentence-concatenate and sentence-insertion knowledge-infusion strategies, the IAN-based DictABSA fails to recognize the correct predictions while the baselines are able to accurately

Table 10
Case Study. The first sub-table shows three ABSA examples, including the sentence, the aspect terms, and the retrieved dictionary knowledge; The second sub-table shows the final predicted sentiment polarities of the ABSA examples by non-BERT-based DictABSA (with IAN) and BERT-based DictABSA (with ABSA-DeBERTa).

ID	Sentence	Aspect term	Dictionary knowledge	Sentiment polarity
S1	The Mountain Lion OS is not hard to figure out if you are familiar with Microsoft Windows.	Mountain Lion OS	A set of programs that control the way a computer system works...	POSITIVE
S2	And the fact that it comes with an i5 processor definitely speeds things up	i5 processor	A machine that processes something...	POSITIVE
S3	Customization on mac is impossible.	Customization	The action of modifying something to suit a particular individual or task...	NEGATIVE

ID	Adopted OriABSA & KI strategies	Baseline	Sent-Concat	Sent-Insert	RNN-based encoder	BERT-based encoder
S1	*IAN (non-BERT)*	NEG ✗	NEU ✗	NEU ✗	NEG ✗	POS ✓
	ABSA-DeBERTa (BERT)	NEG ✗	NEU ✗	NEU ✗	POS ✓	POS ✓
S2	*IAN (non-BERT)*	NEU ✗	NEU ✗	NEU ✗	POS ✓	POS ✓
	ABSA-DeBERTa (BERT)	NEU ✗	NEU ✗	POS ✓	POS ✓	POS ✓
S3	*IAN (non-BERT)*	NEG ✓	NEU ✗	NEU ✗	NEG ✓	NEG ✓
	ABSA-DeBERTa (BERT)	NEU ✗	NEG ✓	NEG ✓	NEG ✓	NEG ✓

predict the sentiment. This error is due to the long-term dependencies problem, that is, the DictABSAs, which adopt RNNs with language sequence infusion strategies, are susceptible to forgetting the previously learned context semantics when learning the extra-long dictionary knowledge.

(iii) A comparison of the different knowledge-infusion strategies shows that the two improved feature encoder infusion strategies (BERT & BiLSTM-Attn encoder-based DictABSAs) overwhelmingly outperform the language sequence infusion strategies (CONCAT & INSERT-based DictABSAs). This is because the feature encoders independently model the external knowledge, which avoids the long-term dependencies problem. Thus, the feature encoder infusion-based DictABSAs obtain better semantic representation vectors to contribute to the aspect term sentiment classification.

(iv) The IAN-based DictABSA predicts the wrong sentiment polarity "NEG" for instance 'S1', when applying the RNN-based encoder as the knowledge infusion mechanism. However, we obtain the correct sentiment polarity "POS" when applying the BERT-based encoder as the knowledge infusion mechanism. This observation verifies our intuition that Transformer BERT has a better semantic representation capability than the traditional RNNs and thus makes little contribution to judging the aspect term sentiment polarity.

7. Discussion

A better understanding of the various attributes of unknown things can help humans make better next-step strategic decisions. Motivated by this real-life phenomenon, we explore the contribution of noun explanatory knowledge from the dictionary in enriching the feature representation of aspects for aspect-based sentiment analysis. Specifically, we leverage the descriptive knowledge extracted from the Oxford Dictionary and we intuitively design several novel knowledge infusion strategies to better incorporate this external knowledge into the original ABSA models.

7.1. Results analysis

In our experiments, several designed knowledge-infusion strategies are combined with many original ABSA models to constitute various dictionary knowledge-enhanced ABSA methods using the adopted module hot-plugging strategy. We call these novel external knowledge enhanced methods DictABSAs. The knowledge-infusion strategies comprise four implementations: *Sentence Concatenat*, which directly concatenates the noun descriptive sentence after the original sentence, *Sentence Insertion*, which inserts the noun descriptive sentence into the position after the target aspect term, RNN-based Encoder, which utilizes an RNN and attention module to independently model the knowledge semantics, and BERT-based Encoder, which utilizes a Transformer BERT encoder to independently model the knowledge semantics. Comprehensive experiments show that the two simple token-level infusion mechanisms do not work well due to the long-distance dependency problem, especially for non-BERT ABSA models. More in-depth experiments demonstrate that RNN-encoder mechanisms and BERT-encoder mechanisms achieve substantial also significant performance gains over all the adopted ABSA baselines. This proves that these strategies solve the long-distance dependency problem and thus can make full use of the superiority of dictionary knowledge. Compared with these ABSA baselines that trained only on supervised datasets, extensive experiments on five benchmarks over ten baselines confirm the overwhelming advantages of our DictABSA in handling the ABSA task. Thus, returning to human common sense is an effective direction for future technological research into artificial intelligence.

7.2. Theoretical and practical implications analysis

By comparing the previous knowledge-enhanced methods that incorporate other types of knowledge such as knowledge graphs (KGs) (Ghosal et al., 2020; Zhang et al., 2020), sentiment lexicons (Fan et al., 2019; Teng et al., 2016; Zou et al., 2018), and syntactic dependency trees (Sun et al., 2019) for ABSA, the major strengths of the proposed DictABSA are highlighted as follows:

- From a theoretical point of view, emphasizing the aspect term's characteristics is a more desirable choice than other various knowledge types for the ABSA task. The sentiment lexicons-enhanced ABSA (Zou et al., 2018) focuses more on enriching the features in the degree of sentiment semantics. Similarly, the KG-enhanced ABSA (Zhang et al., 2020) utilizes semantic-emotion knowledge from KGs, which also essentially focuses on features in the degree of sentiment semantics. Different from these methods, our DictABSA focuses more on enriching the features in the degree of aspect term semantics rather than only in the sentiment semantics. Compared with previous ABSA works, our DictABSA has opened up a new also prospective research perspective for subsequent knowledge-enhanced ABSA research.
- Dictionary knowledge is easier to retrieve through network crawling than other types of external knowledge such as knowledge graphs (KGs) and sentiment lexicons. It is of great significance and great application potential applying the DictABSA in practice. In the case of where there is no dictionary, one can be acquired by simply constructing a generator to summarize the description explanation of a word.
- Moreover, due to the adopted hot-plugging strategy, our DictABSAs are significantly different from other external knowledge-enhanced ABSA methods. Both the ABSA inference module and knowledge infusion module are essentially separated and detachable from each other. The exclusive properties of DictABSA ensure more model assembly possibilities and superior flexibility when incorporating dictionary knowledge into unprecedented new ABSA models.

8. Conclusion and future work

In this paper, we investigate the effect of external knowledge (dictionary noun-explanations) for aspect-based sentiment analysis and thus propose the Oxford Dictionary knowledge augmentation ABSA approach, called DictABSA, which incorporates commonsense knowledge through a joint framework consisting of three modules: DKR, OriABSA and KI. To be specific, the module OriABSA has ten different model implementations in our experiments. Also, we devise four knowledge infusion strategies to take advantage of the superiority of our DictABSA. The results of our extensive experiments on five benchmarks verify the effectiveness and advantage of our proposed external knowledge-enhanced DictABSA. Finally, due to the adopted hot-plugging strategy and the fact that dictionary knowledge is relatively easy to acquire through automatic network crawling, our DictABSA can be applied in practice easily. To encourage more research in this area and to ensure reproducibility, we have publicly released the implementation of DictABSA on GitHub.[20]

In the future, we plan to conduct research into the following problems: (i) Is it possible to improve ABSA performance by leveraging the extracted external knowledge from specific domain-related textual data (e.g. enterprise news, official announcements)? (ii) Can the synonyms, antonyms, and interpretations related to opinion-attitude words be used as external knowledge to improve ABSA performance? (iii) Can dictionary explanation knowledge be used in a useful way through the multimodal learning to achieve better task-related performance together with model interpretation? The dictionary noun explanation is a kind of external knowledge that is readily accessible, so other NLP tasks such as named entity recognition, information extraction, and question answering also are suitable to be infused with dictionary knowledge. For different NLP tasks, practical knowledge-infusion strategies may help researchers better utilize dictionary knowledge. Thus, designing appropriate task-specific dictionary knowledge-infusion mechanisms for other NLP tasks is crucial. This will also be a future research focus of various NLP approaches based on external dictionary knowledge enhancement.

CRediT authorship contribution statement

Weiqiang Jin: Conceptualization, Methodology, Project administration, Software, Resources, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Biao Zhao:** Conceptualization, Investigation, Writing – review & editing, Supervision, Visualization. **Liwen Zhang:** Formal analysis. **Chenxing Liu:** Methodology, Validation. **Hang Yu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

[20] https://github.com/albert-jin/DictionaryFused-E2E-ABSA.

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